

Smart Nation 2.0

A Thriving
Digital Future
for All



MINISTRY OF
DIGITAL DEVELOPMENT
AND INFORMATION





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We thank all the organisations and individuals that have contributed to this report.

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Foreword by Prime Minister

In 2014, Singapore set an ambition to become a Smart Nation. It was anchored on a vision to use technology to improve our citizens' lives.

Ten years on, we have achieved many of our original ambitions. We have developed digital products and services that made everyday life more convenient and comfortable for Singaporeans. Technology was also instrumental in our fight against COVID-19. It helped us to stay connected, even when we had to stay apart physically.

We are now living in a digital landscape that is vastly different from a decade ago. Today, we are more dependent on technology. It has been woven into our daily lives, becoming an integral part of how we live, work and play. Technology is also advancing at a faster pace than before. All these mean that technology will have an even bigger impact on our lives, often in ways that are hard to predict. There will be tremendous opportunities ahead, but also new uncertainties and challenges.

We are therefore refreshing our Smart Nation strategies. In Smart Nation 1.0 we focused on building up capabilities and encouraging the use of technology. Under Smart Nation 2.0, we will sharpen our focus on using technology more effectively to transform our future and shape our nation together.

This report outlines our strategies to achieve three key goals: Growth, Community and Trust. Growth, because technology must empower our people and businesses to reach their full potential. Community, because we must use technology to unify, not divide. And Trust, because Singaporeans must be able to go online with confidence, knowing that their wellbeing is not compromised.

These three goals form the key thrusts of Smart Nation 2.0. To realise them, we have to experiment, innovate and push new frontiers. We have to try new things and help each other along the way. I am confident that, working together, we can create a thriving digital future for all Singaporeans.



We are therefore refreshing our Smart Nation strategies. In Smart Nation 1.0 we focused on building up capabilities and encouraging the use of technology. Under Smart Nation 2.0, we will sharpen our focus on using technology more effectively to transform our future and shape our nation together.



Sincerely,
Lawrence Wong
Prime Minister and
Minister for Finance, Singapore



Foreword by Minister

When Prime Minister Lawrence Wong launched the Forward Singapore exercise to renew our social compact about two years ago, my team and I embarked on a parallel journey. We believed that the Smart Nation initiative, launched in 2014, could contribute tangibly to realising the vision of Forward Singapore. Our exploration would lead us to identify a set of goals for its next bound that serves as a new compass to guide our strategies and actions.

In the past decade, there has been steady progress in each of the four pillars of our Smart Nation. Our digital economy is vibrant and expanding. Our digital society is inclusive and more cohesive than those observed elsewhere. We have progressively transformed public service delivery. We are comprehensively tackling the concerns of digital security. Internationally, our efforts are well regarded.

The Smart Nation (SN) 2.0 journey therefore began by examining how these developments have improved citizens' lives. It acknowledged the gaps that remain and novel opportunities and challenges that have emerged. There is immense pressure globally for governments to act, yet very few ready playbooks to learn from and adapt.

In response, Singapore has evolved our digital governance approach to incorporate five key lines of effort. The first focuses on building up infrastructure, both physical and digital. The second aims to grow digital capabilities in the workforce, enterprises, research and development, and the government. The third recognises that laws and regulations must evolve to both restrain and facilitate digital developments. The fourth involves mobilising the nation to support digital inclusion. The fifth highlights the importance of international cooperation to tackle common concerns given the borderless nature of technology.

Our experience in developing these foundations strengthens our conviction that a narrow focus on digitalisation is incomplete. Instead, we must continuously examine the impact of technology on society and direct digital developments toward outcomes that benefit Singaporeans. In doing so, we are guided by the need to uphold our shared values as a nation to progress together.

With these considerations in mind, we articulated a refreshed vision of our Smart Nation that builds on the foundations we have set. It must be a Smart Nation we can trust, a Smart Nation that helps all of us grow, and a Smart Nation that brings people together in cohesive communities. These must be the goals that unify the many strategies that are part of SN2.0.



We are fortunate that in Singapore, the Ministry of Digital Development and Information has been entrusted with a strong mandate to lead our Smart Nation initiative. ”

For example, the Digital for Life movement rallies the public, private, and people sectors to promote digital inclusion. In our refreshed National Artificial Intelligence (AI) Strategy, empowering workers with AI skills is as important as safeguarding our society against unethical use of AI.

We are also designing SN2.0 as a 'living book' that will evolve as digital developments unfold and our toolkit expands. For example, we will set up a new agency to strengthen online safety. Through a new Fellowship programme, our educators will be equipped with deeper knowledge to reimagine how our students might learn digital skills.

We are fortunate that in Singapore, the Ministry of Digital Development and Information has been entrusted with a strong mandate to lead our Smart Nation initiative. We thank our many partners who have contributed to this national effort. We look forward to working with you to be a Smart Nation that cares for all Singaporeans, giving everyone the opportunity to thrive in our digital future.



Sincerely,

Josephine Teo

Minister for Digital Development
and Information, Singapore

Introduction



In 2014, Singapore launched our inaugural Smart Nation initiative, driven by a compelling vision: **to build better, meaningful, and fulfilled lives for our people, enabled by technology.**

Significant technological developments, such as growing connectivity, cloud computing, and big data, were sweeping the globe. To seize the opportunities brought about by these new technologies and digital tools, we encouraged Singapore to embrace digital transformation across the government, economy, and society.

Ten years on, we have made significant progress. We rank first in the world for digital inclusion.¹ Our digital economy contributed 17.7% to our Gross Domestic Product in 2023, up from 13.8% in 2018.² About 99% of our government services can be completed online. We have used digital technologies to raise citizens' quality of life across domains, from healthcare to education and infrastructure. We have also made daily activities more convenient.

Digital technologies are now ubiquitous. They have become an integral and pervasive part of our lives, influencing how we live, work, and play. Technology still holds tremendous promise. It can help Singapore and Singaporeans reach new frontiers. But if we are to continue harnessing technology as a force for good and maximise opportunities for our people, we must also address risks that have arisen, such as challenges to our individual wellbeing and social harmony.

Ten years on, we are refreshing our Smart Nation vision for the road ahead. This report articulates Singapore's approach to achieving a thriving digital future for all: **Smart Nation 2.0.**

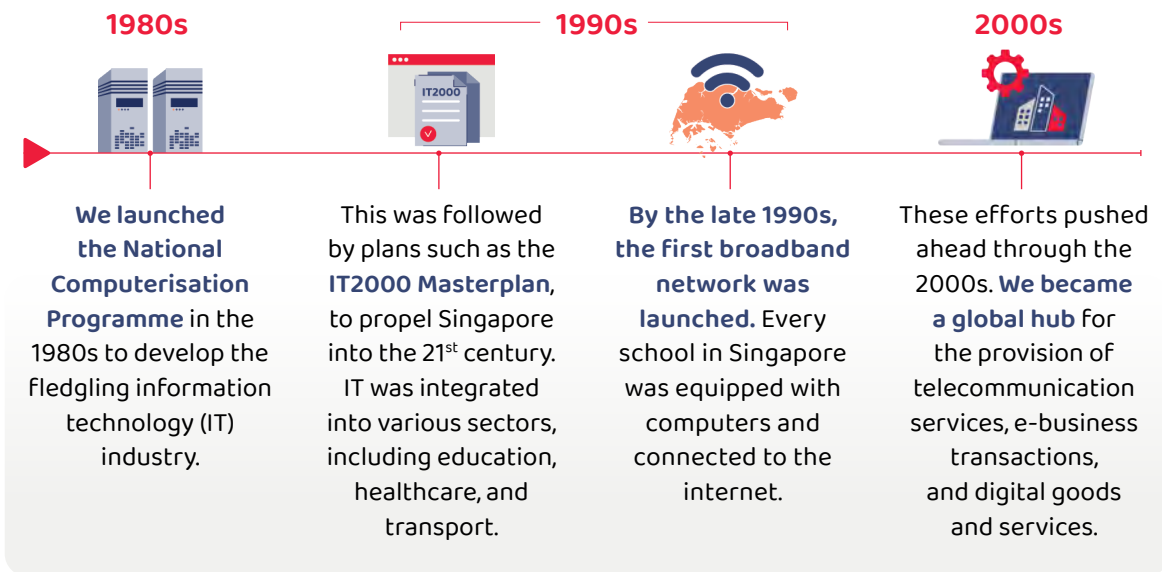
1 Inclusive Internet Index 2022, Economist Impact, 2022.

2 Singapore Digital Economy Report 2024, Infocomm Media Development Authority, end-October 2024.

The Smart Nation 1.0 Journey

Smart Nation 1.0: Decades in the making

When Smart Nation first launched in 2014, it built upon a foundation of digitalisation initiatives that began decades ago.

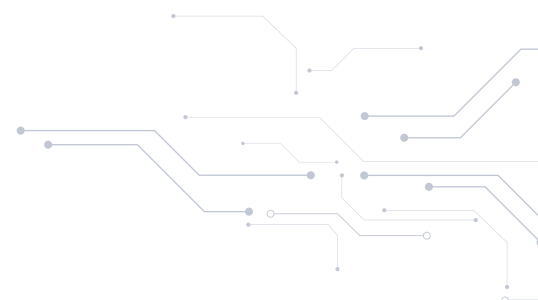


Smart Nation 1.0 was developed to help Singapore ride the wave of change and leverage digital technologies to transform domains like health, transport, education, finance, infrastructure, and environment.

By 2014, the digital landscape had evolved dramatically. The world had become more connected to the internet. Cloud services had gained traction globally. Mobile phone technology had made tremendous leaps.

Smart Nation 1.0 was developed to help Singapore ride the wave of change and leverage digital technologies to transform domains like health, transport, education, finance, infrastructure, and environment.

To increase digital adoption, we launched strategic national projects like our digital identity system, Singpass, and e-payment networks. We promoted digitalisation across the government, society, and economy, and strengthened our digital security.



Introduction

Celebrating Milestones

Over the past decade, we have achieved many milestones. Our investments in digital economy, digital society, digital government, and digital security have borne fruit.



Digital Economy

We aimed to develop a vibrant digital economy and position Singapore as a global hub for innovation and technology, creating new job opportunities. We supported digitalisation programmes for industries and businesses to sustain a thriving enterprise ecosystem.

Digital Economy

95%

of small and medium enterprises (SMEs) adopt digital technology³



Number of tech professionals has grown to **200,000** in 2024, from 160,000 in 2018



95%

of businesses have adopted PayNow and **97%** of Singaporeans are registered on the system



Achieved over **95%** nationwide outdoor 5G standalone network coverage in 2022



Digital Society

We aimed to foster a society where Singaporeans are ready to embrace a digital world. We undertook efforts to improve digital access, improve digital literacy among Singaporeans, and facilitate widespread adoption of technology.

Digital Society

99%

of resident households are connected to the internet as of 2022, up from 88% in 2014



Over **62,000** low-income households have been supported with subsidised broadband and digital devices since 2020

The SG Digital Office has trained more than **340,000** seniors to date and helped more than 11,000 food stallholders adopt e-payment



Digital Government

We aimed to build a government digital to the core that serves with a heart, designing services that are inclusive, seamless, and personalised.

Digital Government

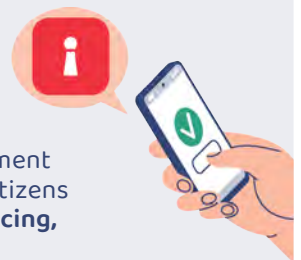
99%

of all transactions between citizens and government can be completed online

High citizen and business satisfaction of **83%**



Extensive suite of digital government products and services to serve citizens and businesses, including **e-invoicing, e-payments, Singpass**, and more



Digital Security

We strengthened our cybersecurity efforts, so that the increasing number of services and transactions that take place online are secure and resilient.

Digital Security

Cybersecurity Strategy

launched in 2016 and refreshed in 2021



[Read more here.](#)

Cybersecurity Act enacted in 2018



[Read more here.](#)

Personal Data Protection Act enacted in 2012, **Public Sector (Governance) Act** enacted in 2018 to maximise value of data in trusted, safe environment

3 Percentage of firms that have adopted at least one digital technology from nine categories of digital technologies (i.e., cybersecurity, cloud, e-payment, e-commerce, data analytics, artificial intelligence, internet of things, blockchain, and immersive media).

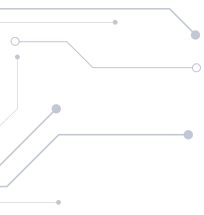
One of the biggest tests for Singapore's Smart Nation initiatives was the COVID-19 pandemic, which unfolded from early 2020. Singapore's economic and social activities went online. Technology became central to Singapore's fight against the virus. The robust digital capabilities that we developed over the years enabled us to respond swiftly and effectively.

- ▶ **We kept Singaporeans safe with digital tools** such as SafeEntry and TraceTogether, which allowed contact tracing to be conducted at scale. We set up and launched a vaccine appointment system in less than a month, which facilitated vaccinations for over 90% of our population in one year.
- ▶ **We kept our economy and society going.** We ensured learning could continue in schools, swiftly shifting to full home-based learning with the use of digital platforms and tools. We were able to disburse economic and social support schemes quickly through online platforms.
- ▶ **We kept Singaporeans up to date with trusted sources of information,** through platforms such as Gov.sg and the GoWhere website. This helped the public collect supplies such as face masks and find clinics.

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Singaporeans from all walks of life stepped up to care for others in the community. Technology amplified their impact. Citizens created videos to teach children about the pandemic. Others raised funds online for those hard-hit by COVID-19. The pandemic proved our mettle and spirit as we used technology to overcome novel challenges as a nation.

This can-do spirit exemplifies our Smart Nation vision. Ultimately, Smart Nation 1.0 was about deploying digital products and services across domains to improve citizens' lives: a focus that will continue to anchor our vision in the future.



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CHAPTER 1

Improving Citizens' Lives



Improving Citizens' Lives

Over the past 10 years, our efforts across digital economy, society, government, and security have created digital products and services that tangibly improve citizens' lives, including in areas like travelling, learning, and enjoying a clean and green environment. Some of these products make daily activities more enjoyable in small but powerful ways; others provide citizens with faster, more personalised services.

We present key examples of such products and services in this section. We also reflect on how we have experimented and adapted, developing products and services iteratively by considering citizens' feedback, learning from setbacks, and pivoting quickly where necessary.

We give a glimpse into some of the next-bound digital products and services that we are creating. Their worth is not measured by how advanced the technology is, but by how much they better our people's lives.

In the years ahead, our Smart Nation vision will remain firmly anchored on improving citizens' lives. As citizens' concerns and needs evolve, we will continue to develop better digital products and services for them.



Improving Citizens' Lives

A. Everyday Activities Made More Seamless and Convenient



i. Transacting

Ten years ago, we took our first steps to enable citizens to make swift and efficient transactions. We launched Fast and Secure Transfers to facilitate direct, real-time financial transactions between consumers and businesses across different banks.⁴ Today, we have an extensive e-payments network that allows people to buy or sell goods and services more conveniently. Citizens can use their phones to buy lunch or shop for groceries through instant payment systems like PayNow.

In addition, citizens can access and spend monies or credits disbursed from government agencies via GovWallet. To date, more than 430 million payouts have been disbursed through GovWallet, including those from schemes like the NS55 Recognition Package, which marked 55 years of National Service.

Singaporeans can also utilise Singpass to verify their identity and securely access over 2,700 services from government transactions to banking and insurance, reducing the need for physical trips to service centres.

This foundation of digital payments and digital identity means that our people can conduct most of their transactions online. Essential transactions are made more efficient and convenient. Indeed, with 99% of government services available online, citizens can transact with the government from anywhere and at any time.



Singaporeans can use Singpass to verify their identity and securely access over 2,700 services.



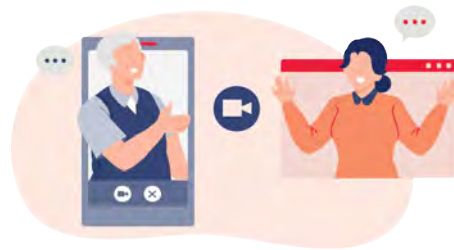
99%
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⁴ Fast and Secure Transfers is an electronic funds transfer service that was launched on 17 March 2014.

Responding to citizens' feedback: SingapoRediscovered Voucher scheme

The SingapoRediscovered Voucher scheme was launched in 2020 to revitalise our tourism sector, which was heavily hit by the COVID-19 pandemic. However, many citizens found the online redemption process too complicated. The Singapore Tourism Board acted swiftly on feedback and made changes to the process, allowing citizens to redeem their vouchers at Community Centres and Residents' Committee Centres. This experience highlighted the importance of making digital initiatives accessible and convenient for all, including citizens who are less familiar with digital services.



ii. Communicating

At least nine in 10 of our residents communicate online, using instant messaging, emails, voice calls, or video calls to stay in touch with friends and loved ones.⁵ This is possible because of our advanced telecommunications infrastructure.

Singapore's investments in our Nationwide Broadband Network (NBN) in 2006 laid the foundation for reliable and high-bandwidth connectivity infrastructure. At the time, broadband speeds were offered at 30 megabits per second. Today, more than 85% of households enjoy broadband plans with speeds of at least one gigabit per second (Gbps). Speeds of up to 10 Gbps are expected when the NBN upgrade is completed in 2026. With higher broadband speeds, citizens can look forward to a more seamless experience when communicating and interacting with their loved ones.

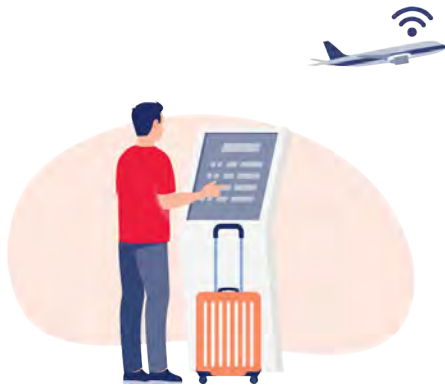


Speeds of up to
10 Gbps
are expected when the
NBN is upgraded in 2026



5 Digital Readiness Survey 2022, Ministry of Digital Development and Information, 2023.

Improving Citizens' Lives



iii. Commuting and travel

Today, Singaporeans plan their daily commute in advance, get updates on bus arrival times, and find the fastest route between places through a variety of wayfinding applications. This is made possible by the Land Transport Authority's DataMall, which collects and makes data across our land transport network available.



Glimpses of the Future

Keeping our public transport system running smoothly

About 7.5 million public transport trips are taken in Singapore every day. We deploy digital monitoring and tools powered by artificial intelligence (AI) to help keep our transport system running smoothly. These tools help identify potential faults and defects before they occur, facilitating efforts to address issues before they cause major disruptions.

We deploy digital monitoring and tools powered by AI to help keep our transport system running smoothly.

We keep our roads flowing smoothly using data and video analytics to identify defects or prioritise when roadworks are needed.

When motorists need to pay for parking spaces, they no longer have to scramble to the nearest petrol station to buy physical parking coupons, estimate how long they need the lot for, or worry about forgetting to display coupons on the dashboard. Vehicle owners can instead use apps like Parking.SG to pay for parking charges on the go.

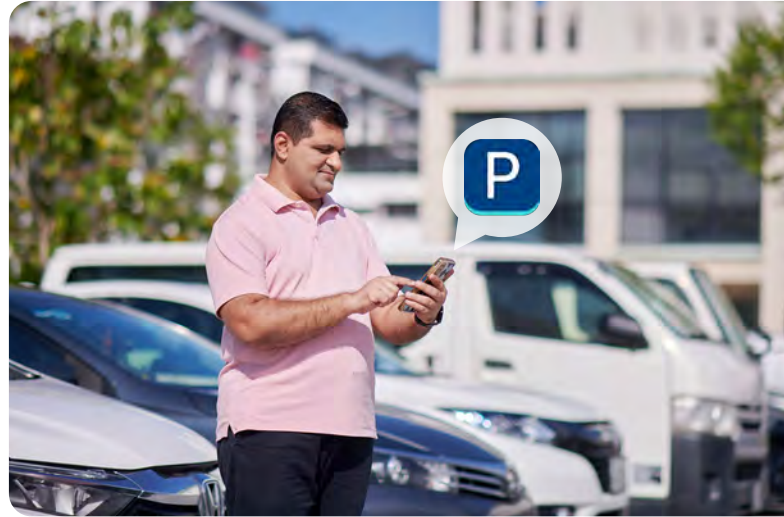


Image Source: Immigration & Checkpoints Authority



Image Source: Immigration & Checkpoints Authority

Many Singaporeans also take pride in our efficient, seamless immigration clearance process when they travel. Automated lanes in the arrival hall make it a joy to return home.

In 2024, Singapore rolled out passport-less immigration clearance at our checkpoints, making us the first in the world to implement it on a large scale. This began with the introduction of quick response code clearance for car travellers at the land checkpoints in March, followed by clearance using facial and iris biometrics at Changi Airport in August.

Soon, the Home Team Science and Technology Agency will also be deploying AI models such as its eXaminer X-ray model to enhance security threat detection at immigration checkpoints.

With these new technologies, Singaporeans can enjoy greater security and convenience when travelling.

Improving Citizens' Lives



iv. Sustainable living

Technology plays an important role in creating smart, sustainable, and liveable environments for Singaporeans. We live in clean and well-managed estates and enjoy recreational activities in beautiful parks, all of which are carefully maintained with the help of technology.

a. Smart, clean homes and neighbourhoods

We use smart technologies to plan, develop, and manage newer Housing & Development Board (HDB) towns to create better living environments for residents. For example, sensors promptly inform town councils of faults in estate services such as water pumps and lifts, enabling quick responses.

Citizens can also report municipal issues conveniently to Kaki (the OneService Chatbot)⁶ or the OneService app, playing their part to maintain a clean, hygienic, and safe neighbourhood. AI is used to swiftly route residents' feedback to relevant agencies, which then provide updates on the issue through the platform.

Glimpses of the Future

Smarter homes, smarter districts

Technology can help us manage our districts more efficiently and sustainably.

HDB and the Agency for Science, Technology, and Research developed an advanced modelling tool called the Integrated Environmental Modeller (IEM) to carefully plan newer estates such as Northshore in Punggol and Tengah. The IEM uses high-resolution 3D models to simulate how environmental factors such as wind flow and air temperature might affect the urban landscape. This optimises the design and placement of HDB blocks and communal spaces to create cooler neighbourhoods. The IEM also identifies ideal spots for solar panels.

We have also launched Singapore's first smart district: the Punggol Digital District (PDD). It features an Open Digital Platform (ODP) system built by the Government Technology Agency of Singapore (GovTech) and Jurong Town Corporation. The ODP seamlessly integrates various systems within PDD, such as those controlling electricity, water, and lighting, enabling them to communicate with each other in real time. For instance, the ODP automatically adjusts air-conditioning and lighting levels during periods of low human traffic within buildings in the district, facilitating more efficient and sustainable management of commercial and industrial buildings.



Image Source: Housing & Development Board

6 Available on WhatsApp, Telegram, and Instagram.



Image Source: National Environment Agency

A spirit of experimenting, iterating, and learning

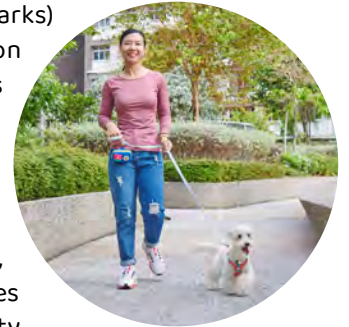
In using technology to improve citizens' lives, we embrace a mindset of experimentation, iteration, and continuous learning. Initiatives such as Hack for Public Good exemplify this approach. It is an annual month-long hackathon where officers at GovTech's Open Government Products (OGP) team brainstorm, develop, and prototype digital solutions to address problems faced by the public. Many ideas have emerged from this hackathon; some have gained traction and progressed to the implementation stage. This process of iteration has yielded valuable insights.

Equally important is the willingness to stop or pivot when things do not work out, especially when technology has advanced or community needs have evolved. In 2017, we trialled the Lamppost-as-a-Platform initiative to assess the viability of using lampposts to mount sensors. We closed the project after determining that a large-scale rollout would not be cost-effective when compared to other methods, such as mobile sensors powered by fuel cells, which emerged as a viable alternative while the project was underway.

b. Enjoying our city in nature

Many Singaporeans enjoy outings to neighbourhood parks and recreational areas. We keep our beautiful city in nature flourishing with the help of technology.

The National Parks Board (NParks) carefully tends to six million trees across the island. Its Remote Tree Management System uses AI and LiDAR (Light Detection and Ranging) technology to create a digital twin of Singapore's urban trees, alerting NParks officers to issues such as changes in tree stability.



The National Environment Agency has also successfully trialled an autonomous waterway cleaning machine at Sungei Seletar to remove flotsam and keep our waterways clean. The machine will be deployed in other regions by the end of 2024.

Glimpses of the Future

Trees.sg: deepening our human connection with nature

We are connected to the environment in more ways than we imagine. In Singapore, many of us have good memories of the trees that line familiar roads. Think of the saga seeds we used to collect in school or the leafy rain trees that provide shade over our park benches.

Trees.sg is a website developed by NParks and GovTech that aims to map every tree in Singapore. We can identify the species, age, and interesting traits of any tree, down to the smallest shrubs outside our houses or grand old heritage trees that are hundreds of years old. We can even send trees a virtual hug!

Improving Citizens' Lives



Enjoying an active lifestyle, supported by technology



Madam Nancy Tan, who is 66 this year, is constantly on the move: working, attending her Zumba classes, or doing household chores. She uses the Healthy365 app and the Health Promotion Board's fitness tracker to record her step count as she goes about her day. She also uses the app to book piloxing classes and earn 'Healthpoints' when she buys healthier groceries, food, and drinks.

"When I track my exercise and lifestyle habits on the app, I feel motivated to stay active and take care of my health."

– Nancy Tan –



v. Healthy living

Digital technologies can empower Singaporeans to lead healthier and more active lifestyles. They enable families to enjoy recreational activities together and support seniors to keep active as they enter their golden years. We have developed innovative products and platforms for our healthcare sector to enhance our service delivery and citizens' wellbeing.

a. Active lifestyles

Digital tools like the Healthy365 app help seniors to keep track of their health and fitness goals.

b. Advancements in healthcare

We use technologies such as AI to improve our healthcare service delivery and patient wellbeing in areas like clinical diagnosis and the management of hospital operations.



Glimpses of the Future

Using AI to ease pain during childbirth

Singapore uses AI to assist doctors with the precise delivery of spinal anaesthesia during childbirth. Typically, doctors would have to manually use their hands to identify the landmark for spinal needle insertion.

The KK Women's and Children's Hospital (KKH) developed the world's first AI-powered ultrasound-guided identification system to improve this process, raising the success rate of delivering spinal anaesthesia by 22%. The precise spinal needle insertion at first attempt can improve the quality of anaesthesia and reduce complications such as nerve irritation for the patient.

Following its implementation at KKH in May 2023, this system will be introduced to other hospitals such as Singapore General Hospital, National Neuroscience Institute, National University Health System, and Tan Tock Seng Hospital.



Image Source: KK Women's and Children's Hospital

Improving Citizens' Lives



vi. Lifelong learning

Just as technology can help us keep fit and healthy, it can also keep our minds active and engaged. We have used digital technologies to make knowledge come alive for all Singaporeans and support them in their lifelong learning journeys.

a. Supporting our children's education

As children progress through school, parents may find it challenging to keep up with a multitude of tasks: tracking when their children's enrichment and co-curricular activities are held, signing consent forms for trips, and more. The Ministry of Education's (MOE) Parents Gateway app consolidates this information online, enabling parents to read and respond on the go. Parents can even book appointments for parent-teacher meetings on the application.

The **Parents Gateway** app enables parents to read and respond on the go



Glimpses of the Future

Making online content simpler for students

Online content is typically written in language pitched at the reading level of adults and may present challenges for students with literacy barriers. This can be discouraging for students and make them reluctant to read.

Simpler is an AI-powered browser plugin which allows students to access online content at their preferred reading level. Students can select any text in their browser and simplified versions will appear on a side panel for their convenience. Simpler also helps define and explain technical terms. With this plugin, students can focus on learning rather than struggling to decipher text.

Simpler is currently undergoing a pilot test involving over 110 students taking up G1 English Language.⁷ Initial feedback has been promising, with teachers and students describing its use as a seamless, enjoyable learning experience.

Future iterations of Simpler could offer even more personalised, supportive learning environments for students and possibly simplify texts in languages other than English.

⁷ Throughout their secondary school education, students can take subjects at three subject levels, known as G1, G2, and G3, mapped from the previous Normal (Technical), Normal (Academic), and Express standards respectively. Students have flexibility to adjust their subject levels at appropriate junctures, based on their strengths, interests, and learning needs.

b. Enjoying our libraries

Many Singaporeans cherish a visit to the nearby library where they can immerse themselves in a world of books, stories, and learn something new. Through technology, we can extend these unique experiences digitally for more citizens to enjoy.

Initiatives under the National Library Board's (NLB) LAB25 (Libraries and Archives Blueprint 2025) leverage technology to reimagine libraries and archives in Singapore. For example, StoryGen allows users to put their own twists on well-loved stories and see them come alive in an AI-powered multimedia experience. This prototype service was first launched at the Central Public Library and is currently roving at other libraries.

Libraries also provide an inviting environment for Singaporeans to learn and apply new digital skills; users can pick up information literacy skills and experiment with technologies like 3D printing.

Citizens can also access library resources while on the move. With the NLB mobile app, Singaporeans have convenient access to a growing range of digital resources, such as e-books and online courses.



Image Source: National Library Board

AI-enabled multimedia and human imagination come together in StoryGen, where library visitors make their own stories come alive.



Initiatives under the National Library Board's LAB25 leverage technology to reimagine libraries and archives in Singapore.



Improving Citizens' Lives



vii. Caring for one another

Technology can help us better care for our loved ones and the needy.

a. Caregiving

Digital apps like HealthHub support those who need to take care of their loved ones. With their consent, caregivers can view loved ones' health records such as vaccination history and test reports online. They can also create and track appointments on these platforms.

Caregivers can use the care services recommender function on the SupportGoWhere website to identify schemes and services that they may need, including mental health and financial assistance initiatives.

b. Protecting seniors in need

We use technology to support seniors who live independently, keeping them connected to their communities.



Help at all hours for seniors through wireless alerts

To provide timely help to seniors in need, HDB and GovTech will expand the provision of wireless alert alarm systems to households with at least one senior aged 60 and above and living in about 170 rental blocks islandwide. Seniors who press the alert device in their homes will be connected to CareLine, a 24/7 hotline that responds to those in distress and provides round-the-clock emergency response services. This initiative is expected to benefit nearly 27,000 seniors.





A personalised care ecosystem for independent seniors

Social service agency Lions Befrienders has created a personalised digital care ecosystem—the I-OK Ecosystem—to support the independence and safety of the seniors they serve.

Lions Befrienders has distributed 1,000 IM-OK tablets that seniors can use to let their next-of-kin, Lions Befrienders' staff, or volunteers know that they are well by tapping the 'I'm OK' button on the device at scheduled times. The tablets can also send reminders to seniors to take their medications.

One such user is Madam Jackie Cheng, a 64-year-old retiree who lives alone and benefits from the daily check-ins through the IM-OK tablet.

One morning, Madam Cheng experienced severe vertigo which left her too dizzy to get out of bed. As she missed her check-in on the IM-OK device, a notification was sent to her nephew, prompting him to visit her and take her to the doctor. Madam Cheng has since recovered well.



Image Source: Lions Befrienders Service Association (Singapore)

Senior beneficiaries learning how to use the IM-OK device.

c. Supporting one another in times of need

Digital technologies can be deployed to mobilise citizens to support one another during emergencies.



Enabling swift emergency assistance

Many cardiac arrest or minor fire cases that the Singapore Civil Defence Force (SCDF) receives may be quickly attended to by the public in the vicinity.



Anyone can register as a Community First Responder (CFR) via SCDF's myResponder app. The app alerts CFRs to nearby cases of cardiac arrest and minor fires. A chat function allows CFRs to communicate with the SCDF Operations Centre to get real-time guidance.

Since myResponder's launch in 2015, over 180,000 CFRs have registered themselves and 80 lives have been saved by CFRs who attended to cardiac arrest cases through the app.

Improving Citizens' Lives

B. Smart Nation 2.0: A Refreshed Vision

We have made significant progress in how we use technology to overcome challenges and improve lives. Our investments across digital economy, society, government, and security have paid off and raised Singapore's international standing. Several global indicators recognise Singapore as a leader in areas like digital infrastructure and inclusion.

The landscape of digital technologies is evolving rapidly. Generative AI tools have progressed from completing text paragraphs to generating realistic videos and writing code. On the horizon, emerging technologies like quantum computing have the potential to usher in a new era of technological advancement.

Such developments present a world of new opportunities for Singapore and for humanity.



- ▶ **Augmented or virtual reality** can help connect people across distances and times.



- ▶ **AI** is already enabling us to deliver more personalised, tailored services to citizens. It could accelerate drug discovery for the healthcare sector and spur new treatment options for patients.



- ▶ **Quantum technology** could be used to secure our communication channels against malicious interception of data. It could help us better understand our world and propel breakthroughs in fields like physics, chemistry, and materials science.

However, digital developments have also given rise to new challenges.



- ▶ **Online harms** such as scams, cyberbullying, misinformation, and disinformation challenge our sense of safety and wellbeing.⁸ Cybersecurity incidents and digital outages have also led to significant financial losses and disruptions.



- ▶ **Technology-induced isolation,⁹ mental health concerns, and online polarisation** strain our social fabric. Digital exclusion can also worsen socio-economic inequalities and deepen social divides.



- ▶ **Workers worry about job disruptions** and their impact on livelihoods. Some feel that technology is changing too quickly and in ways that do not benefit them.¹⁰

Our refreshed Smart Nation vision must address these needs and concerns. We still believe that technology can be harnessed for good. We must continue to seize new opportunities and reap the benefits that digital developments can bring.

But we need to go beyond a narrow focus on digitalisation. We must continuously examine the impact of technology on our humanity and society. We must shape and direct digital developments toward outcomes that benefit Singaporeans. We must always be guided by the need to uphold our shared values as a nation to move forward together.



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⁸ Annual Scams and Cybercrime Brief 2023, Singapore Police Force, 18 February 2024.

⁹ Singapore Perspectives 2024 Pre-Conference Poll, Institute of Policy Studies, 29 January 2024. The poll found that young people aged 21 to 34 were more likely to report higher levels of social isolation and loneliness, with 53% of respondents finding it easier to talk to people online than offline.

¹⁰ 2024 Edelman Trust Barometer, Edelman Trust Institute, 14 January 2024.

Improving Citizens' Lives

C. Smart Nation 2.0: Our Vision and Compass for the Future

Smart Nation 2.0 seeks to achieve three key goals: **Trust, Growth, and Community**.

We aspire to be a Smart Nation that we can trust, a Smart Nation that helps us grow, and a Smart Nation that keeps us together.

More than a destination, these goals serve as our compass to shape how we use technology to improve citizens' lives and create a thriving digital future for all. They also reflect key areas of feedback and concern that citizens have raised over the years, gathered through surveys, research, and other feedback channels.¹¹



¹¹ These include surveys such as the Digital Readiness Survey, Online Safety Poll, and consultations with citizens and industry partners to seek feedback on key policy initiatives such as the Digital Enterprise Blueprint.



Our **Smart Nation 2.0** strategies are akin to a '**living document**' that will be continually iterated and adjusted as digital developments unfold.

We will adapt our approaches over time in close consultation with citizens and businesses, responding to advancements in technology and emerging challenges. The goals of Trust, Growth, and Community will guide our responses to digital developments, even as our strategies evolve.



CHAPTER 2

Trust



Trust

Singapore takes a holistic approach to building a Smart Nation that is safe, reliable, and secure. Our citizens and enterprises must be able to depend on our digital infrastructure. Harmful online activities should be detected and disrupted quickly, with effective help and redress readily available to victims. Our citizens must be equipped with knowledge and skills to go online safely and confidently.

While the government takes the lead in developing policies, providing support and resources, and rallying partners to achieve these objectives, a collective effort is still needed. This requires individuals, groups, and organisations to do their part.

To build a Smart Nation that we can trust, we will:

A.



**Enhance
the security and
resilience of our digital
infrastructure**

B.



**Strengthen
our fight against
harmful online
activities**

C.



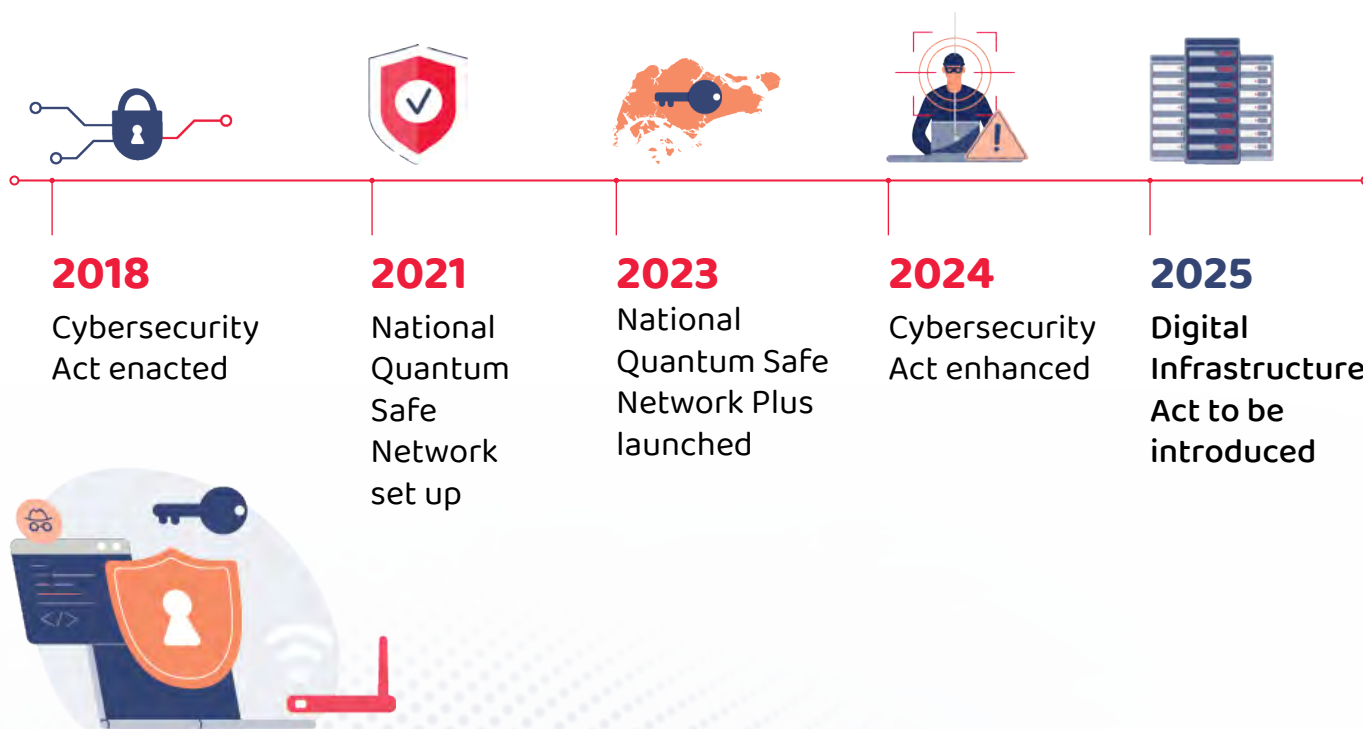
**Expand
our trusted
digital spaces
together**

A. Enhancing the Security and Resilience of Our Digital Infrastructure

Digital infrastructure forms the foundation of our Smart Nation. It powers our digital economy and the digital products and services that citizens rely on. As with physical infrastructure like utilities and public transport, digital infrastructure such as broadband and mobile networks, data centres, and cloud services need to have high availability. Unexpected disruptions can greatly impact individuals and businesses and erode trust and confidence in the use of digital technology.

For example, in 2024, a flawed software update by cybersecurity firm CrowdStrike caused major disruptions at multiple locations around the world. In 2023, a four-hour data centre outage caused by the failure of a cooling system resulted in the widespread disruption of banking services in Singapore.

Highly reliable digital infrastructure and services give people and businesses peace of mind and allow us to make plans with certainty. While we cannot completely prevent disruptions or cyberattacks from occurring, our key systems must be well-protected against them, with our people and organisations prepared to respond and recover.



i. Ensuring security and resilience of organisations and systems

Over the years, Singapore has put in place comprehensive measures to uphold the security and resilience of key organisations and their IT systems.

- ▶ The 2018 Cybersecurity Act fortifies the security of computer systems that deliver essential services like banking, energy, and water supply. Owners of critical information infrastructure must implement cybersecurity measures, conduct regular risk assessments and audits, report incidents, and participate in cybersecurity exercises.
- ▶ Digitalisation has since become more pervasive. Threats have increased both in scale and impact. In response, we enhanced the Cybersecurity Act in 2024 to further safeguard our critical information infrastructure and regulate a broader range of organisations and systems. These include systems at high risk of cyberattacks for limited time periods and services foundational to our economy and way of life, such as cloud services and data centres.



We intend to introduce a new Digital Infrastructure Act in 2025 to improve the resilience and security of

key digital infrastructure and services that can cause significant impact to the economy and society if disrupted. The proposed law will address broader security and resilience concerns faced by digital infrastructure and service providers, including technical misconfigurations and physical hazards such as fires and cooling system failures. This complements existing cybersecurity regulations, helping to reduce the likelihood and mitigate the impact of systemic disruptions.

Amidst evolving technologies and threats, a trusted Smart Nation is one that anticipates future needs and proactively meets them.

- ▶ Quantum computing will spur innovation and bring benefits. But there are also risks that the technology could be used to break the encryption methods we currently rely on for securing our communications and transactions. In 2023, we launched the National Quantum Safe Network Plus (NQSN+). Building on over a decade of local research and trials, NQSN+ supports network operators in deploying nationwide networks that are secure against attacks from quantum computers, so that businesses and organisations can safeguard their critical data.





Keeping a Smart Nation running smoothly

Data centres play a key role in our digital economy and Smart Nation. They manage, process, and store data, enabling large-scale digital applications and services that we all depend on.

Security and resilience are top-of-mind concerns for data centre service providers like ST Telemedia Global Data Centres (STT GDC). To ensure uninterrupted service and to better manage cyber and physical risks, STT GDC provides for power and cooling redundancies in their centres. This allows them to perform maintenance work without impacting customers. The systems and processes they have in place for security and resilience have also been validated with industry certifications, including for information security management and data centre design. Fire and other emergency drills are performed on a regular basis as part of broader efforts to develop and test their incident response, business continuity, disaster recovery, and crisis communication plans.

Behind the scenes, data centre staff work to meet present needs and prepare for the future. Jafred Choo started his journey with STT GDC through an internship placement facilitated by the Singapore Institute of Technology. He is now a Senior Team Lead responsible for ensuring round-the-clock service delivery and for his team members' training and development.

“Demand for high-performance computing and data services is set to grow. Our team has a forward-thinking mindset, continuously updating our know-how and developing solutions to deliver quality services that businesses and society can depend on. We are proud to contribute to Singapore's growth and success.”

– Jafred Choo –



ii. Building capacity, fostering collaboration

Most organisations today need digital security and resilience regardless of their size. In Singapore, eight in 10 enterprises encountered at least one cybersecurity incident in 2023. Failure to address risks upfront exposes organisations to preventable disruptions.

In this regard, Singapore is most concerned about small and medium enterprises (SMEs) and supports them in three ways:¹²

- ▶ Helping SMEs improve their cyber hygiene.
- ▶ Raising the cybersecurity standards of digital systems used by SMEs.
- ▶ Working with sector leads to customise implementation guides.

At the same time, stakeholders (industry, academia, and governments) need to cooperate closely to address complex threats and risks. We are expanding and deepening collaborations to improve monitoring and detection, build capacity, and better defend against cyber threats.

- ▶ We are establishing a new National Cybersecurity Command Centre at the Punggol Digital District to better coordinate cyber defence operations, strengthen collaboration with industry and academia, and spur cybersecurity innovation.
- ▶ The Cyber Security Agency of Singapore (CSA) has forged partnerships with leading technology companies such as Microsoft, Google, and Dragos. It has also established bilateral cooperation mechanisms with the United States of America, the United Arab Emirates, the Republic of Korea, and a wide range of international partners.

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¹² Digital Enterprise Blueprint, Ministry of Digital Development and Information, 29 May 2024.

B. Strengthening Our Fight Against Harmful Online Activities

Malicious online activities can cause grave harms. Scams proliferated through online channels could result in financial loss and emotional distress. Violent or sexually explicit content damages victims' sense of safety and wellbeing. False or misleading online information can threaten social cohesion.

There have been increasing efforts to address such harms. But perpetrators are constantly changing their tactics. Accordingly, we have to continuously update our responses.

 In our efforts to combat scams, we have taken a holistic approach: improving operations, implementing strong countermeasures, and adopting new technology. 

i. Bolstering our capability to detect and disrupt harmful activities

Harmful online activities can unfold rapidly and with little warning. Scammers can siphon away victims' life savings within minutes. Falsehoods can go viral and are extremely difficult to debunk. Artificial intelligence (AI) tools may present new threats, making scams or manipulative campaigns more effective and pervasive. Holding perpetrators to account is a challenge as they may be anonymous or operating out of jurisdiction.

We need to better detect and disrupt such activities even before harm is caused. In our efforts to combat scams, for example, we have taken a holistic approach: improving operations, implementing strong countermeasures, and adopting new technology.



Continual improvements in combatting online scams



Improving our operations

An effective scam response involves detecting unfolding scams, tracing proceeds, and preventing scammers from communicating with victims and moving funds.

The Singapore Police Force (SPF) set up the Anti-Scam Command (ASCom) in 2022, where bank staff and government officers work side by side, fighting scams together. In 2023, ASCom prevented S\$44 million in potential scam losses, froze more than 19,600 tainted bank accounts, and recovered more than S\$100 million.

We have also updated our laws to allow SPF to act more effectively against money mules and negligence leading to the misuse of bank accounts or Singpass credentials for scams or other criminal activities.¹³

Implementing strong countermeasures

Working together, public agencies and industry partners have developed new safety features to secure communication networks, preventing scammers from reaching potential victims and funds.

Since July 2024, all government agencies have been sending SMSes from a single gov.sg sender ID, making it easier to distinguish authentic government SMSes from fraudulent ones.

To prevent scammers from reaching potential victims through SMS messages, the Infocomm Media Development Authority (IMDA) established the Singapore SMS Sender ID Registry. Registration is mandatory for all organisations that use SMS sender IDs. In the first three months of mandatory registration, scam cases involving SMSes fell by 70%.

Telcos have acted against calls from known scam numbers, spoofed local numbers, and robocalls, helping to block over 390 million potential scam calls in 2023. Subscribers can also choose to block all international calls and SMSes on their mobile phones.

Banks now have emergency 'kill switches' for customers to suspend compromised accounts quickly and the 'money lock' feature lets customers prevent sums from being transferred digitally.



Image Source: Open Government Products

¹³ In 2023, we amended the Computer Misuse Act 1993 and the Corruption, Drug Trafficking, and Other Serious Crimes (Confiscation of Benefits) Act 1992.

✍️ We can use the enhanced ScamShield app to check if content is scam-related and make scam reports more easily. ✍️



Adopting new technology

Even as scammers abuse technology to cause harm, we must use technology to fight back. Partnerships and continuous improvement are essential to this.

In 2024, we worked with Google to create a new fraud protection feature under Google Play Protect for all Android devices in Singapore. This feature blocks risky sideloaded apps, protecting users against malware-enabled scams. In six months, it blocked nearly 900,000 attempts to install high-risk apps.

We enhanced the ScamShield app. The previous version delivered good results, blocking over 178,000 phone numbers, sender IDs, and website links believed to be used for scam communications. Now, in addition to filtering calls and SMSes, users can use the app to check if suspicious messages, phone numbers, and website links are scam-related and make scam reports more easily, including by uploading screenshots.

We strengthened authentication processes for banking services. Since July 2024, banks have been phasing out One-Time Passwords (OTP) for bank account logins by customers using digital tokens, removing the risk of disclosing OTPs to scammers. Banks will also be introducing stronger authentication methods for online banking services to further disrupt phishing and malware scams.



We also invest in technology that helps us better understand online harms and develop improved solutions. For example, we invested S\$50 million to build the Centre for Advanced Technologies in Online Safety (CATOS), bringing researchers, companies, and practitioners together to build technological capabilities for a safer internet.

- ▶ Together with the Ministry of Health Office for Healthcare Transformation, CATOS uses emotion analysis technology to support a youth-led online community mental health forum, aiming to flag content that may require human moderators' attention.
- ▶ CATOS partners with Checkmate, a local community group running a WhatsApp-based service that checks publicly submitted messages and images for misinformation and scams. CATOS will augment Checkmate's volunteer-led efforts with a tool that lets Checkmate respond more quickly and accurately.

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Online Safety



Fulfilling the need for speed

Chloe Lim, a Senior Product Manager at the Government Technology Agency of Singapore's (GovTech) Anti-Scam Products team, oversees the recursive Machine-Learning Site Evaluator (rMSE). It identifies potential scam websites based on various technical attributes, as well as similarities to known scam sites.

From its beginnings as a research project, rMSE has grown significantly through Chloe and her team's efforts. Today, rMSE can evaluate hundreds of thousands of suspicious sites a day. Each site is evaluated in two to three seconds, with accuracy rates of well over 90%. SPF uses rMSE as one of its indicators to evaluate potential scam websites found or referred through police reports, ScamShield, and other channels. Local internet service providers and Google's Safe Browsing service are alerted to malicious sites and access to them is swiftly disrupted. This helps citizens to avoid visiting such sites and falling prey to scams.

"Scammers frequently change their tactics and methods in unexpected and sophisticated ways. Technology is a critical force multiplier that enables us to work more effectively and efficiently against them. We will keep on improving and experimenting with new solutions to protect ourselves and our loved ones."

– Chloe Lim –



ii. Protecting Singapore through laws and regulation

Online harms are constantly evolving. To keep pace, our laws and systems for policing crime, ensuring justice and fairness, and providing help and recourse must also be regularly updated.

a. Moving with speed and agility

With few playbooks or proven solutions for digital governance globally, we are often pioneers in designing and implementing protections for citizens and businesses. We adopt an accretive approach—moving quickly, thoughtfully, and in close consultation with citizens and key stakeholders.

- ▶ We were among the first countries in the world to introduce a dedicated Cybersecurity Act in 2018 to regulate owners of critical information infrastructure and cybersecurity service providers, and the Protection from Online Falsehoods and Manipulation Act in 2019 to counteract the effects of online falsehoods.
- ▶ The Personal Data Protection Act 2012 was amended in 2020 to strengthen organisational accountability and consumer protection, while giving greater assurances to organisations using personal data for innovation and other legitimate purposes.

We were among the
first
countries
in the world to
introduce a dedicated
Cybersecurity Act



- ▶ More recently, we strengthened our safeguards against harmful online content such as advocacy of suicide and self-harm through the Online Safety (Miscellaneous Amendments) Act 2022 and against criminal online content or activities through the Online Criminal Harms Act 2023. Under these laws, online services must implement preventive and mitigating measures. They must also comply with government-issued directions to prevent access to egregious content and disrupt criminal activities.

With few playbooks or proven solutions for digital governance globally, we are often pioneers in designing and implementing protections for citizens and businesses.



We will continually review our laws to respond effectively to new threats such as abuses of deepfake technology.

- ▶ Through the Elections (Integrity of Online Advertising) (Amendment) Bill, we propose new measures during elections against digitally generated or manipulated online election advertising that realistically depicts a candidate saying or doing something that he or she did not in fact say or do.
- ▶ To foster a safe information space, we will also introduce a new Code of Practice to require specified social media service providers to prevent and counter abuse of digitally manipulated content on their services.

b. Enhancing online safety and assurance

Those affected by online harms have important and legitimate needs that should be supported. These include putting a stop to ongoing harmful activities, holding perpetrators to account for harms suffered, and seeking counselling or medical help to manage serious mental and emotional hurt.

We have enhanced our systems and laws over the years to better support these needs. We established the Protection from Harassment Courts where people can file legal claims for harassment, doxxing, false statements, and other similar activities. Volunteers under SPF's Victim Care Cadre programme also offer psychological and emotional support to scam victims during investigations, directing them to professional resources for longer-term support.

Victims, experts, and community organisations that we consulted have called for swifter and more effective ways for victims of online harms to seek help and recourse.



To enhance online safety and provide stronger assurance to victims of online harms, **we will set up a new agency for online safety and assurance, backed by a new law to improve protection for victims of online harms.**

- ▶ The agency will act on behalf of victims to direct perpetrators and online service providers such as social media services to put a stop to the online harm quickly, providing added assurance on top of existing regulatory levers and court-based processes available today. We will focus on the most serious and prevalent online harms affecting victims in Singapore such as cyberbullying and non-consensual sharing of intimate images.
- ▶ The new law will also better enable victims to seek civil remedies against perpetrators of online harms.



Supporting victims of online harassment

SHECARES@SCWO (Singapore Council of Women's Organisations) is Singapore's first support centre for victims of online harms. Recognising that the victims' needs regarding help and recourse can vary from person to person, the centre provides phone and text helplines, counselling services, and pro bono legal assistance. It also works closely with the police and internet platforms with case and in-app reporting.

Jane (not her real name) approached SHECARES@SCWO in early 2024. An unknown person was posting photographs of her on social media, along with highly damaging and unsavoury allegations. The person also offered to sell information about her residential address and later posted her phone number on social media.

Distressed by these events, Jane confided in a friend and learnt about SHECARES@SCWO. She immediately reached out to the centre for assistance. Her main concern was that friends and family would see these posts which would affect how they see and relate to her. The immediate priority was to get the posts taken down.

Saira Sultanah, a counsellor at the centre, provided counselling and emotional support and reported the content to the social media service provider. With the benefit of the centre's status as a trusted flagger, the service provider quickly took down the content. Jane was very grateful for the centre's assistance and relieved that no further posts were made thereafter.

"We take a client-centric and trauma-informed approach, assuring clients that what they face is not something to be taken lightly and that help is available. Our partnerships with internet platforms have been useful in the timely removal of distressing content. This gives clients quicker relief, allowing them to focus on their mental health journey or on seeking other forms of recourse. We have various services available and clients can decide on their preferred approach. Everyone's journey is different and no harm is too small to seek help for."

– Saira Sultana –



C. Expanding Our Trusted Digital Spaces, Together

We increasingly gather and interact in shared online spaces. These spaces may be chatgroups, forums, social media platforms, games, or other shared virtual environments. We are all part of digital communities and networks.

Building trustworthy online spaces is a collective effort. While laws and regulations deter and punish harmful, negative conduct, we also need to encourage everyone to make good choices and act responsibly online so that responsible use of technology becomes the norm. We can only foster and expand trusted spaces online if all of us protect and respect one another's wellbeing. Our aspiration must be that our online spaces lift spirits, foster connections, and are conducive to our wellbeing.



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Based on a 2024 survey,
3/4
of Singaporeans
have encountered
harmful online
content



Increasingly, citizens must have the mindsets, skills, and tools to make responsible decisions on technology use and online safety. These include recognising online deception and manipulation, practising good cyber hygiene, knowing where and how to seek help, and engaging responsibly online.

- ▶ Based on a 2024 survey, about three in four Singaporeans have encountered harmful online content such as cyberbullying content, content inciting racial or religious tension, and violent content.¹⁴
- ▶ Another study in 2022 by Nanyang Technological University's Centre for Information Integrity and the Internet showed that seven in 10 Singaporeans have unknowingly shared fake news articles.

¹⁴ Online Safety Poll conducted by the Ministry of Digital Development and Information in April 2024.

i. Educating and empowering everyone

We have implemented programmes and provided resources to help citizens navigate online spaces safely and confidently.

- ▶ IMDA launched the Digital Skills for Life framework in 2024 to help Singaporeans improve their digital skills. Bite-sized learning resources on basic digital competencies, including being safe and discerning online, are available on their website. In-person coaching and workshops are also available at various community hubs.
- ▶ The National Library Board's (NLB) S.U.R.E. (Source, Understand, Research, Evaluate) programme promotes good information searching and discernment practices. The 'I can ACT against scams' campaign by SPF and the National Crime Prevention Council builds awareness and confidence among the public on practical steps they can take to better protect themselves against scams.

We will strengthen our public education and empowerment initiatives to help us deepen our knowledge and skills and keep up with new developments.



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▶ Strengthening collaboration

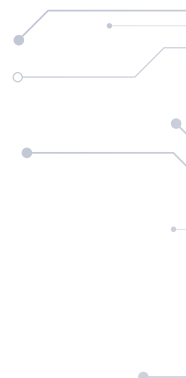
Partnerships between public, private, and people sector organisations can combine different strengths and greatly enrich our programmes and initiatives. The S.U.R.E. Learning Community, set up in June 2024 by NLB and CheckMate, regularly conducts meetings and workshops on information literacy topics and skills. Members can put their skills into practice on the CheckMate platform or as S.U.R.E. Ambassadors, allowing more people to directly contribute to protecting our community and loved ones from misinformation and scams.

▶ New ways of learning

We are studying the use of innovative and personalised learning approaches to help more citizens gain and apply skills to safeguard their wellbeing, including improving Singaporeans' resistance to deepfake-enabled manipulation and deception.

▶ Keeping updated on new features

Digital products and services regularly introduce new wellbeing and safety features, such as digital watermarking to identify AI-generated content. But poor awareness and understanding can lead to low adoption or worse, leave users vulnerable to bad actors by creating a false sense of security. Our public education and empowerment initiatives must keep up with such new features and developments, so that we can benefit from them while understanding their limitations and avoiding their risks.



ii. Inspiring action

Achieving digital wellbeing requires collective long-term commitment. We need to foster a culture where we treat one another well, build kinder online networks and spaces, and safeguard positive norms and values of online interaction.

Across the people, private, and public sectors, various organisations have strived to create safe, healthy, and positive digital experiences for our children and youth. This is a group we will pay special attention to.

Creating safe and positive digital environments for our children and youth



Our children and youth should be able to use digital devices in healthy ways and navigate online spaces safely. We want them to experience the wonders of digital technology and use it to enrich their lives, without causing them mental distress or harm.

Healthy device use

There is growing understanding and awareness that excessive screen use negatively affects the brain development of young children and takes time away from physical play and face-to-face interactions, which are important to a growing child. Time limits should be set and children under 18 months should not use screens unless for video chatting. It is important to have a plan to balance screen use with other activities such as family time and physical activities. The government will continue to review the latest evidence and update its guidance on child screen use.



Fostering healthy device use takes concerted and consistent effort. Parents can access relevant guidelines and tips through online portals like Parent Hub, Families for Life, and Digital for Life. Our pre-school curriculum also creates children's learning experiences with digital technologies that are meaningful, appropriate, and carefully considered to promote the safety and wellbeing of children.¹⁵

Healthy online environments

For older children and youth, their online environments should be age appropriate and conducive to their safety and wellbeing.

► Enhanced regulations on service providers

Under the Code of Practice for Online Safety, designated social media service providers need to have differentiated content standards and measures for young users. These include more restrictive account settings for children and adolescents, as well as tools for parents and children to minimise the impact of and exposure to harmful or inappropriate content.

We are working closely with app distribution service providers to develop a new Code of Practice. This code will require designated app stores to protect children and adolescents, possibly including measures for effective age assurance. This will reduce their risk of exposure to inappropriate and harmful apps, even as they access a wide range of digital content and services online.

► Tailored resources for parents, teachers, and youth

The Ministry of Education (MOE), the Ministry of Social and Family Development (MSF), and the Health Promotion Board have developed a Parenting for Wellness toolbox to equip and empower parents to build strong parent-child relationships, strengthen their children's mental wellbeing and emotional resilience, and support their children's digital journey while keeping them safe online. The toolbox is being made available in phases from September 2024. The first tranche of resources covers various topics such as managing screen use, addressing cyberbullying, and developing good digital habits.

The Ministry of Digital Development and Information is also working with MOE and MSF to develop a Positive Use Guide on Technology and Social Media, expected to be released in 2025. This authoritative, research-informed resource will help guide the healthy and positive use of technology and social media and provide recommendations on how to mitigate negative impacts.

MOE will continue providing and improving cyber wellness resources for students, including within the Singapore Student Learning Space and as part of Character and Citizenship Education.

We will work with partners to develop a suite of resources to help individuals support those being harmed online and equip parents, teachers, and social workers to support children and youth who have suffered online harms.

¹⁵ Teaching and Learning Guidelines on the Use of Information and Communication Technology in Pre-School Centres, Ministry of Education, 2017; Nurturing Early Learners Framework, Ministry of Education, 2022.

Google, IMDA, and the Media Literacy Council have helped train over

158,000

parents and children since 2022 on safe and responsible actions online



► **Dedicated partnerships and programmes**

There is strong momentum among the public, private, and people sectors to jointly develop programmes to help our youth and children go online safely and confidently.

CSA has collaborated with Microsoft to use Minecraft Education to offer workshops to primary and secondary school students. These engaging workshops aim to raise awareness on cybersecurity and encourage adoption of good cyber hygiene practices. Launched in September 2024, students learn how to protect themselves against cyber threats such as phishing and malware through immersive scenarios and relatable messages within the Minecraft game. At the end of the workshops, students create unique cyber worlds on Minecraft and compete in an annual hackathon.

Google's Be Internet Awesome campaign, in partnership with IMDA and the Media Literacy Council, has helped train over 158,000 parents and children since 2022 on safe and responsible actions, fundamentals on protecting privacy, and addressing cyberbullying and misinformation. These efforts are further enhanced by the Interland game that helps children learn through play. Educators can also download lesson plans and resources to teach about digital safety, including through classroom activities that bring the lessons to life.



Keeping our children safe online

Daryl Sung and Melissa Goh are parents who participated in Focus on the Family Singapore's webinar on 'Screenwise Parenting' as they were concerned about their two primary school-going girls' screen time and device usage habits.

Through the webinar, they learnt how parents should set clear and safe boundaries for screen time and online interactions and what areas of concern to be mindful of.

Since the webinar, they have sought to apply what they have learnt through regular conversations with their children on what they encounter online.

"The knowledge gained from the webinar puts us in a better position to educate our children about the digital world and to navigate it together with them."

– Daryl Sung and Melissa Goh –



CHAPTER 3

Growth



Growth

Digitalisation has been a significant contributor to Singapore's economic and wage growth and can create new opportunities for our people.

Of every S\$100 value-added in our economy, at least S\$17 can be attributed to digital-related activities. Our digital economy amounted to S\$113 billion in 2023, larger than the financial services and insurance sector and comparable with the manufacturing sector.¹⁶

Today, there are more than 200,000 tech professionals in Singapore who earn median wages higher than the rest of the workforce.¹⁷ Although they represent around five percent of all jobs, there are thousands more jobs across all sectors that have been enhanced by digital technologies.

With technologies like artificial intelligence (AI), we can potentially push the boundaries of productivity even further and strengthen the competitiveness of our industries.

Technology can also help us uncover new knowledge and possibilities. In the realm of research, new technologies can help expand the limits of human discovery. We can grow our understanding of the world and of humankind so generations after us can live fuller and richer lives.

But technological advances may cause unwelcome disruption to the economy and labour markets. Citizens worry about job displacement and keeping up with the pace of change. We must make every effort to help our workers and enterprises adapt while developing technology to augment our capabilities.

Growth is therefore about preparing our people to thrive with technology in a future markedly different from today. Each successive generation will confront new realities and must keep learning new skills to navigate an increasingly digital world.

To build a Smart Nation that helps us grow, we will:

A.



**Power our
digital economy**

B.



**Empower our
enterprises and workers**

C.



**Prepare our
next generation**

¹⁶ Singapore Digital Economy Report 2024, Infocomm Media Development Authority, end-October 2024.

¹⁷ Based on median wages earned by resident tech professionals in Singapore, compared with median wages of the resident workforce.

A. Powering Our Digital Economy

In 2023, our digital economy contributed 17.7% to our GDP.¹⁸ In the last five years, it has achieved a compounded annual growth rate of 11.2%, nearly double the 5.8% growth rate for Singapore's overall economy.¹⁹ Our digital economy will remain a strong engine of growth in our economy for many years to come.

Our digital infrastructure—the invisible foundation that powers the growth of our digital economy—must continue to be strong and future-ready. At the same time, we must develop a robust response to the global challenge of sustainable data centre growth. This will involve catalysing innovations and promoting the adoption of energy efficient technologies and low-carbon energy alternatives.

We must stay at the leading edge of technology by harnessing the transformative potential of AI. In 2023, we launched the National AI Strategy (NAIS) 2.0 which set out our vision and plans to achieve AI for the Public Good, for Singapore and the World. We have since made good progress, boosting broad-based AI adoption and partnering with leading technology players to build capabilities and know-how in Singapore.

As we continue these efforts, we will also strengthen the use of AI for scientific research to help us expand the frontiers of human discovery right here in Singapore.



Prime Minister Lawrence Wong launching the National AI Strategy 2.0 at the Singapore Conference on AI in 2023.

¹⁸ Singapore Digital Economy Report 2024, Infocomm Media Development Authority, end-October 2024.

¹⁹ Based on nominal growth rates.

i. Future-proofing our digital infrastructure

Robust, resilient, and future-ready digital infrastructure is the foundation upon which a strong digital economy can grow. It enables reliable internet connectivity and high-speed data transfer that supports a full range of economic and social activity.

Singapore is recognised for being at the forefront of digital infrastructure development. We are consistently ranked among the best countries in the world for digital connectivity and infrastructure.²⁰

One of our pivotal initiatives was the Next Generation Nationwide Broadband Network (NBN) which provides high-speed broadband to all homes, offices, and schools across Singapore. We decided to develop the NBN in the mid-2000s, well before mobile internet applications became widespread. This early investment positioned Singapore to reap the benefits of then-budding services like e-learning and online collaboration when demand took off.

Furthermore, the NBN prepared us to weather unexpected crises. When the COVID-19 pandemic struck, demand for digital services surged as more activities quickly moved online. Our high-speed broadband network kept Singapore going.

We will provide capacity to

double
submarine cable
landings within the
next 10 years



We will anticipate needs, stay ahead of trends, and continue to upgrade our digital infrastructure to preserve Singapore's leading position. Through implementing the Digital Connectivity Blueprint, we will:

- ▶ Provide capacity to double submarine cable landings within the next 10 years to strengthen digital connectivity between Singapore, Southeast Asia, and the rest of the world. This will potentially catalyse S\$10 billion in overall submarine cable investments or more.
- ▶ Invest up to S\$100 million in our NBN to enable broadband speeds that are 10 times faster (up to 10 gigabits per second) within the next five years.
- ▶ Ensure best-in-class resilience and security of our increasingly critical compute infrastructure.
- ▶ Promote greater adoption of the Singapore Digital Utility Stack in order to facilitate a wider range of digital transactions and foster interoperability.²¹

²⁰ Digital Cities Index 2022, Economist Impact, 9 June 2022; IMD World Digital Competitiveness Ranking 2023, IMD World Competitiveness Center, 30 November 2023.

²¹ The Singapore Digital Utility Stack refers to the public soft digital infrastructure stack, which contains four baseline foundational digital utilities—digital identity, e-payments and e-invoicing, document attestation, and data exchange—that enable broad-based digitalisation.

As we expand our digital infrastructure capacity and connectivity networks to meet growing demands, we will be steadfast in upholding our climate commitments and push the boundaries of environmental sustainability in the digital domain.

- ▶ In 2023, Singapore launched one of the world's first standards for optimising energy efficiency for data centres in tropical climates.
- ▶ This was followed by the launch of the Green Data Centre Roadmap in May 2024 to chart a sustainable growth pathway for data centres.

ii. Extending our frontiers with AI

AI is a strategic national priority. In 2019, we launched our first NAIS that outlined plans to embark on national AI projects in areas such as education and healthcare.

However, advances in AI—which have increased in accessibility, scale, and impact—necessitated a review of our strategies. This led to a refresh of NAIS in 2023 which seeks to maximise value creation and empower individuals, businesses, and communities to use AI with confidence, discernment, and trust. We will invest more than S\$1 billion over the next five years to support these efforts.

A vision for more sustainable data centres



Senior Minister of State Janil Puthucheary launching the Green Data Centre Roadmap at Asia Tech x Singapore 2024.

Data centres are an essential component of our digital infrastructure stack. Singapore is a regional data centre hub, with a total capacity exceeding 1.4 gigawatts. Our operational data centre capacity per capita exceeds that of countries such as China and the United Kingdom.²² But data centres also consume significant amounts of energy and we must carefully manage the resource footprint of data centre growth.

The Green Data Centre Roadmap sets out plans to provide at least 300 megawatts of additional capacity in the near term and more through green energy deployments. Through this additional capacity and partnerships with industry, we aim to seed innovation for sustainable data centre growth. This includes:

- ▶ Accelerating data centre energy efficiency through best-in-class technologies to maximise efficiency and compute capacity.
- ▶ Accelerating data centres' use of green energy to expand capacity.

22 Driving a Greener Digital Future: Singapore's Green Data Centre Roadmap 2024, Infocomm Media Development Authority, 30 May 2024.

The AI ecosystem in Singapore is growing more vibrant by the day. We are seeing greater domestic adoption of AI in public service delivery, from predicting hospitalisation times to smoothening border clearance, and in industry, from improving customer experience to combatting fraud.

We are building on this momentum to scale new peaks of excellence across multiple sectors while minimising the risks and harms that AI brings.

a. Transforming key sectors

AI has the potential to shape how sectors organise, operate, and serve their customers more efficiently and effectively. We intend to establish sectoral AI centres of excellence to help transform key economic sectors and boost Singapore's economic competitiveness.

- ▶ We launched our first sectoral AI centre of excellence for the manufacturing sector in September 2024. It aims to develop globally impactful, AI-enabled solutions that address use cases within the manufacturing sector, such as enhancing quality assurance and optimising manufacturing processes. This pilot will inform our approach for subsequent sectoral AI centres of excellence.

We will also attract companies to set up AI centres of excellence to anchor their innovation capabilities in Singapore. Among other measures to support these companies, we will facilitate their access to high-performance compute that is essential for AI innovation and deployment.

- ▶ A good example is American Express which expanded its Decision Science centre of excellence in Singapore, incorporating AI and machine learning (AI/ML) to optimise its customer marketing and services.
- ▶ Local companies are also investing in this space. Singapore Airlines is expanding its generative AI team in Singapore to reimagine customer experiences, improve operational efficiency, and increase workforce productivity.

We partner leading technology providers. For example, in May 2024, Microsoft unveiled its AI Pinnacle Program with five pioneering organisations to accelerate business and sectoral transformation through co-innovation of AI use cases and capability building.

We launched our
first
sectoral AI centre
of excellence for the
manufacturing sector
in September 2024





How DBS Bank uses AI/ML to improve productivity and encourage growth

Anticipating threats and disruptions to the banking sector, DBS Bank has invested heavily in AI/ML, scaling up over 350 use cases which have improved many business functions. These applications have helped unlock additional value for the bank.

DBS is also experimenting with generative AI, with a focus on augmenting employees' work by automating routine tasks. This allows employees to focus on more strategic and value-added activities, such as building deeper customer relationships.

DBS adopts a holistic approach towards AI, focusing on three key areas: processes, technology, and people. The DBS AI Protocol provides staff with a knowledge hub and a set of best practices to standardise the delivery of AI/ML solutions. DBS also developed Advancing DBS with AI, a platform housing over 5.3 petabytes of data with analytics capabilities to enable rapid building and deployment of AI/ML solutions. With the help of these tools, DBS realises value from AI/ML solutions in two to three months, compared to 12 to 15 months previously.

DBS has also invested heavily in upskilling its employees, training over 9,000 of them in AI and data skills. It established a 'data chapter' with 700 data and analytics professionals, providing them with opportunities to deepen their domain knowledge and share their insights with business and support units.

"It is a privilege to be at the forefront of these latest advancements, including the generative AI wave, and to also be a part of a forward-thinking organisation. My colleagues and I have deepened our domain skills with the latest tools, processes, and skills, have opportunities to broaden our exposure across different business challenges, and have a well-defined path for career progression. There is a palpable sense of excitement and energy at the prospect of shaping the future and making a meaningful difference in the industry."

– Matthew See –

Regional Head,
Risk & Business Analytics, DBS Transformation Group



b. Striving for research excellence

We have consistently invested in AI research, having committed more than S\$500 million through AI Singapore under the Research, Innovation, and Enterprise 2020 and 2025 plans.



We will add to our investments in using AI to push the limits of scientific discovery with an additional investment of up to S\$120 million in AI for Science.

- ▶ This initiative will focus on the development and adoption of AI methods and tools that are transferable across multiple domains of science to enhance research productivity and advance scientific discovery. It will also fund collaborations between AI researchers and scientific domain experts in thematic areas that are of interest to Singapore such as advanced materials research and biomedical and health sciences.
- ▶ By bringing together experts in AI and scientific research, the AI for Science initiative will keep our research community competitive and vibrant, putting innovation in service of Singapore's national imperatives and economic development.



AI research enables more personalised healthcare

Professor Wynne Hsu is the Lead Principal Investigator on project Jarvis.²³ Jarvis aims to improve the healthcare delivery process for patients with 3H conditions: Hyperglycemia (diabetes), Hypertension (high blood pressure), and Hyperlipidemia (high cholesterol). Working with clinicians, the research team developed a suite of AI products that facilitate the early identification of patients with higher 3H risks and develop tailored treatment plans for them. These products consolidate multiple functions onto one platform, including patient engagement and screening tools.

“The work we have done through this project is just the beginning and it highlights the importance of collaboration between researchers and clinicians. Working together, we hope to develop more personalised AI tools that are responsive to the needs of our population, leading to better health outcomes for all Singaporeans.”

– Wynne Hsu –



²³ Collaborators of the Jarvis project include the NUS Institute of Data Science, the Singhealth Health Services Research Centre, the Singapore National Eye Centre, the National Heart Centre Singapore, and the Duke-NUS Medical School.

c. Creating a safe and trustworthy environment for AI development and deployment

While novel AI applications are unlocking many opportunities, they also carry risks. Left unmitigated, these risks may cause enterprises and citizens to distrust the technology over time. Our push to promote AI adoption and research should therefore be accompanied by good AI governance.

Singapore seeks to address AI's risks through principled approaches, practical tools, and inclusive international engagement. This is to help our people use AI with confidence and enjoy the benefits it brings.

- ▶ We will regularly review the adequacy of our laws and regulations to ensure that they remain fit for purpose.
 - » In some cases, existing laws and regulations can already adequately address some of the harms associated with AI.
 - » In other cases, we will consider introducing new legislation. For example, the Workplace Fairness Legislation that will be introduced in late 2024 will hold employers accountable regardless of how bias comes about during the hiring process. If AI-enabled hiring systems result in discriminatory employment decisions, they can be reported to the Tripartite Alliance for Fair and Progressive Employment Practices which will work with employers to ensure that such systems uphold the principles of workplace fairness.

- ▶ We will provide and advance sensible, pragmatic guidelines to shape AI developments, as well as promote the development of practical tools to help organisations evaluate their AI models and applications for risks.

- » In 2019, we introduced the Model AI Governance Framework to provide practical guidance on key ethical and governance issues when deploying traditional AI solutions. In 2024, we extended this effort with a new framework which also covers generative AI.
- » To promote the translation of principles laid out in the frameworks into action, Singapore launched the AI Verify testing framework and software toolkit in 2022, making it open-source in 2023 through the AI Verify Foundation. This enables developers and users to demonstrate responsible AI in an objective and verifiable manner that consumers can trust.
- » Building on this, we created the AI Verify Project Moonshot in 2024, one of the first tools in the world which brings together benchmarking, red-teaming, and testing baselines to manage the deployment risks of large language models.
- » We are working with industry and international partners on our Guidelines and Companion Guide on Securing AI Systems.



B. Empowering Our Enterprises and Workers

It is a given that technology changes the way we work and transforms opportunities for our workers. We will remain steadfast in ensuring that growth and progress are shared widely with our workers and enterprises. We must work to use technology to augment our people's capabilities regardless of their starting points and circumstances.

We will help our enterprises take full advantage of technology to raise productivity, transform, and serve customers better. Our suite of initiatives is updated continuously to remain relevant to enterprises at different stages of development.

We will also empower our workers to adapt, succeed, and realise their aspirations in a changing economy. We have built up robust training infrastructure to support all workers in their efforts to upskill, reskill, and remain competitive. We have enabled locals to take up more than 70% of the tech jobs today and will continue to expand Singapore's tech talent pool.²⁴



We must work to use technology to augment our people's capabilities regardless of their starting points.



Senior Minister of State Tan Kiat How launching the Digital Enterprise Blueprint at Asia Tech x Singapore in May 2024.

24 Singapore Digital Economy Report 2024, Infocomm Media Development Authority, end-October 2024.

i. Expanding enterprises' digital capacity

Enterprises, big and small, are the backbone of our economy. Their ability to effectively harness the power of digital technologies will determine how far and fast our digital economy grows.

We have paid particular attention to helping our small and medium enterprises (SMEs) navigate their digital transformation journey. Over the years, we have introduced many initiatives to support SMEs that strive to build stronger digital capabilities:

- ▶ Through the SMEs Go Digital programme, IMDA works with other agencies to develop Industry Digital Plans that serve as digitalisation roadmaps for SMEs across various sectors.
- ▶ The Chief Technology Officer-as-a-Service platform provides SMEs with access to curated digital solutions and consultancy support.
- ▶ Through the GoBusiness platform, enterprises can easily access government e-services and resources, such as licence applications and information on grants.

Our SMEs have benefitted from these initiatives and made good progress in expanding their digital capacity, with nine in 10 enterprises adopting at least one digital technology today, compared to seven in 10 in 2018.²⁵

In 2024, we launched the Digital Enterprise Blueprint (DEB) to catalyse deeper digital transformation. Seven partners have come on board and pledged their support. They are: the Singapore Business Federation, the Singapore Computer Society, SGTech, Amazon Web Services (AWS), Google, Microsoft, and Salesforce.

For the broad base of SMEs, our current initiatives focus on:

- ▶ Promoting the adoption of emerging technologies, especially AI-enabled solutions.
- ▶ Enabling faster scale-ups through cloud-based and integrated solutions.
- ▶ Strengthening cyber resilience through better access to relevant expertise and resources.

Our SMEs made good progress in digitalisation, with

9 in 10

enterprises adopting at least one digital technology today, compared to 7 in 10 in 2018



²⁵ This refers to firms that have adopted at least one digital technology out of nine categories: cybersecurity, cloud, e-payment, e-commerce, data analytics, AI, internet of things, blockchain, and immersive media.

For digitally mature enterprises, customised generative AI solutions offer new promise.

- ▶ IMDA has partnered tech companies such as AWS, Google, and Microsoft to help digitally mature enterprises deepen their understanding of generative AI and access relevant expertise to start innovating.
- ▶ The Digital Leaders Programme will also support ambitious enterprises seeking to become digital leaders in their industries.

For start-ups, we will create a vibrant ecosystem by launching accelerator programmes such as the Ignition AI Accelerator developed by Digital Industry Singapore, Nvidia, and Tribe, a leading start-up ecosystem builder.

Digital tools help businesses expand



Auntie Anne's, a business specialising in freshly baked pretzels, used to face operational roadblocks due to its manual and time-consuming finance workflow. Sales figures captured across its 15 outlets were disorganised. Manual reconciliation and reporting were challenging. There was also no real-time status update on its key business indicators.

Since it adopted cloud-based accounting software, Auntie Anne's easily automates reconciliation of its accounts across all its outlets. It also gained a bird's eye view on its business performance with real-time statistics. Time needed for accounts settlement has been reduced by 60%. The reduction in mundane data entry and reconciliation work has led to improved staff morale, which resulted in lower turnover and greater job satisfaction.



Image Source: Auntie Anne's

ii. Equipping workers with skills to succeed

Embracing digital technologies is crucial for staying relevant and competitive. Workers must be prepared to continually acquire new digital skills that could open new opportunities for growth and advancement.

We recognise, however, that rapid changes in the digital landscape can be overwhelming and daunting for workers. We are committed to helping workers develop the confidence and skills to navigate digital transformation so that everyone moves forward together in our digital future.

Our workers are in fact in a strong position to benefit from further digital developments. Today, Singapore's workforce is highly skilled and digitally confident. The number of tech professionals has grown from 166,000 in 2018 to around 200,000 in 2024, driven by demand from both Information & Communications (I&C) and non-I&C sectors.

We will continue to provide comprehensive training support through programmes designed to help workers upskill and reskill, ensuring that they remain competitive, resilient, and relevant.

Number of tech professionals
has grown to around

200,000

in 2024 as compared to 166,000
in 2018 across all sectors



- ▶ We have trained more students in our Institutes of Higher Learning (IHLs) for the tech sector. In 2023, 8,600 students were enrolled in Information & Digital Technology courses in our autonomous universities, polytechnics, and the Institute of Technical Education (ITE), a 40% increase from the annual enrolment in 2017.²⁶
- ▶ Programmes under IMDA's TechSkills Accelerator (TeSA) enable professionals to acquire in-demand digital skills. Since 2016, TeSA has placed and trained nearly 18,000 locals in tech domains such as cybersecurity, network and infrastructure, cloud computing, software and applications, data and AI, and 5G.
- ▶ To strengthen ITE and polytechnic graduates' employment outcomes and provide multiple career pathways, we have set up the TeSA for ITE and Polytechnics Alliance (TIP Alliance) which aims to provide 1,000 tech job opportunities for ITE and polytechnic graduates by 2025. The TIP Alliance also launched the Skills-Based Hiring Movement to bring about a shift in industry hiring practices toward focusing on skills rather than qualifications.

We are placing strong emphasis on training in AI to prepare our workers for new growth opportunities in the digital economy. Our workforce has demonstrated great interest in adopting and acquiring AI skills. According to LinkedIn, Singapore's AI skills diffusion rate of 20 times is significantly higher than the global average of eight times.²⁷

²⁶ Intake admin data for IHLs.

²⁷ Future of Work Report: AI at Work, LinkedIn, August 2023. Diffusion rate refers to the share of members who have added AI skills to their profiles as of June 2023 compared to January 2016.

- ▶ We aim to triple our AI practitioner pool to 15,000 over the next five years.
 - » We will invest over S\$20 million in the next three years to enhance AI practitioner training for students, covering AI-related SG Digital scholarships and overseas internships in AI roles.
 - » Beyond students, we will support recent graduates and mid-career workers who wish to become AI practitioners by providing them with upskilling opportunities and industry-relevant training through a scale-up of TeSA.
- ▶ We have established strong partnerships with tech companies like AWS, Google, Microsoft, Oracle, and SAP Labs, who have committed to help Singaporeans develop AI skills.

We will also equip the broader workforce to be confident AI users. Through the Jobs Transformation Maps (JTM), we will identify job roles affected by various technologies including AI and provide useful signposts for employers, unions, and workers so they can plan for job redesign or training.

- ▶ Under the I&C JTM that was launched in 2023, we have curated about 200 AI and AI-related courses together with five partners including the National University of Singapore and NTUC Learning Hub.
- ▶ Since August 2023, about 5,000 tech workers from over 800 companies have benefitted from JTM curated offerings including AI courses.



From audit to AI: Deepening digital skills

Mukundraj Raman, also known as Raj, started work as an audit associate after completing his undergraduate degree in Accountancy, with a second major in Accounting Data and Analytics. During the COVID-19 pandemic, he joined Lithan Academy as an adjunct instructor, training mid-career professionals in digital skills.

Raj subsequently moved to a local bank, where he was employed as a data analyst. There, he fell in love with programming and was encouraged to chart a new career path in data science. With the help of the SG Digital Scholarship, he pursued a Master of Science in Data Science at New York University to deepen his skills. In the future, Raj hopes to push for greater adoption of AI across the fintech sector.



C. Preparing Our Next Generation

We will prepare our next generation to expand their horizons and realise their aspirations in a digital future. This goes beyond preparing them for work in a digital economy. We need to prepare our children for life in the digital age.

We are already living in a world where technology is part of everyday life. For younger Singaporeans, understanding how technology works must become as foundational as understanding the natural world.

We want young Singaporeans to be curious, confident, and creative users of technology. Students need to know how best to use technologies to augment their capabilities. Equally, our children must learn how to use technology ethically and responsibly. Instead of becoming over-reliant on it, they must carefully examine its impact to ensure that humans and humanity remain central to technology's development and use.

We will enhance programmes, increase resources for schools, and strengthen support for educators to design effective learning experiences that develop students' digital proficiency.



We have come a long way since 1997, when we introduced the first Masterplan for Information and Communications Technology in Education and committed to providing every five students with one computer. Today, under the National Digital Literacy Programme launched in 2020, all secondary school students have school-prescribed personal learning devices.

Beyond improving access to digital technologies, we have enhanced the teaching and learning of digital and AI-related skills.

Under the National Digital Literacy Programme launched in 2020,

ALL
secondary school students have school-prescribed personal learning devices



- ▶ We deepened the development of computational thinking in the Secondary Mathematics curriculum from 2020, and revised the Lower Secondary Science syllabus to help students develop a better understanding of emerging technologies like AI.
- ▶ From 2026, upper secondary students will have the option of taking up Computing at G1 and G2 levels, in addition to G3 Computing.²⁸
- ▶ Schools have programmes and resources in place to allow students to experiment with technology, such as digital making activities in Applied Learning Programmes.²⁹

When students progress to IHLs, they continue to acquire and develop critical digital skills.

- ▶ All IHLs have introduced modules to raise the baseline digital competencies of students in areas like computational thinking, quantitative reasoning, and digital wellbeing and ethics.
- ▶ IHL students learn about how AI might be used in their fields of study. Those who wish to pursue AI in-depth can also do so via specialised courses or programmes such as ITE's Higher Nitec in AI applications, Nanyang Polytechnic's Diploma in Applied AI and Data Analytics, or Nanyang Technological University's new Turing AI Scholars Programme.

²⁸ Throughout their secondary school education, students can take subjects at three subject levels, known as G1, G2, and G3, mapped from the previous Normal (Technical), Normal (Academic), and Express standards respectively. Students have flexibility to adjust their subject levels at appropriate junctures, based on their strengths, interests, and learning needs.

²⁹ Digital making refers to activities that involve creating things with digital tools, ranging from 3D printing to robotics.

i. Enhancing students' digital and AI-related skills

Building on these foundations, we will continually update and improve plans to ensure that our children have the tools and skills to succeed in tomorrow's world.

After four successive masterplans, the Ministry of Education (MOE) launched the Transforming Education through Technology Masterplan 2030 (EdTech Masterplan 2030) in 2023. It aims to strengthen the development of students' digital skills and leverage technologies such as AI to better customise and personalise learning for every child.

Through the EdTech Masterplan 2030, we will increase the number of digital and AI-related self-paced modules which are hosted on the Singapore Student Learning Space accessible to all students. We will review and refresh existing content to keep them relevant.



We will also introduce new AI for Fun modules. These five- to 10-hour elective modules are available under the Code for Fun (CFF) programme,

which all primary and secondary schools can offer. The CFF was first introduced in 2014 to MOE primary and secondary schools. It exposes students to computational thinking through coding, inventive thinking through digital making, and emerging technologies. More than 50,000 students participate every year.

- ▶ The new modules will build upon the 10 hours of CFF that are already mandatory for primary school students and optional for secondary students.³⁰ They will offer students opportunities for hands-on exploration and tinkering with technology such as through the design of prototypes incorporating AI.



The Edtech Masterplan 2030 aims to strengthen the development of students' digital skills and leverage technologies such as AI to better customise and personalise learning for every child.



³⁰ While CFF remains an optional programme at the secondary school level, almost two-thirds of secondary schools offered the programme in 2023.



Making learning about AI fun for young students

The new AI for Fun modules, covering topics like natural language processing and generative AI, have been piloted at 82 primary and secondary schools.

Northland Secondary School offered the modules under their Applied Learning Programme. Students used generative AI to develop ideas and learnt to programme microcontrollers with object and facial recognition capabilities. Teachers found that the modules helped to deepen students' AI literacy and adaptive thinking skills.

Waterway Primary School offered the modules as part of their Makers Roadmap which teaches students computational thinking, design thinking, and digital making. Students programmed robots to respond to gestures and objects identified through computer vision while gaining a better understanding of the benefits and risks of AI.

"I like how the lessons allow me to explore the different programmes and apps with AI capabilities. It was an eye-opening experience for me and I enjoyed the lessons very much!"

– Eva Chee –
Waterway Primary School student



"The AI function was impressive, allowing students to train the machine themselves. This hands-on experience helps them appreciate AI better by understanding how it works behind the scenes."

– Mohammad Hazmie Jaafar –
Waterway Primary School teacher



ii. Focusing on our educators

As we introduce more learning opportunities for students, our educators play a critical role in inspiring our students to embrace the potential of technology and navigating the risks involved.

As technology advances, educators need up-to-date insights on technological developments, their implications on society, and their impact on teaching or learning. This helps them to better understand their students' needs in digital skills and cyber wellness development.

Educators need up-to-date insights on technological developments, their implications on society, and their impact on teaching or learning. This helps them to better understand their students' needs in digital skills and cyber wellness development.



To support our educators, we will launch a new Smart Nation Educator Fellowship in 2025.

The six-month long Fellowship will help Senior Specialists and Teacher Leaders³¹ better understand the latest digital issues, offering them practical insights on the real-world applications of technology.

- ▶ The Fellowship will also include opportunities for educators to learn from industry experts and participate in hands-on workshops by partners such as the Science Centre Singapore. Educators can then translate knowledge and skills acquired from the Fellowship into curriculum design or effective teaching and learning strategies which they can share with other educators.
- ▶ This Fellowship complements existing professional learning support offered by the National Institute of Education and the Academy of Singapore Teachers, such as out-of-classroom learning opportunities through the Teacher Work Attachment Plus programme and learning journeys.
- ▶ We are confident that this initiative will help passionate educators leverage digital technologies to bring the classroom alive and nurture a confident next generation that will take our Smart Nation forward.

³¹ Senior Specialists help to shape the national curriculum. Teacher Leaders include Master Teachers, Senior Teachers, and Lead Teachers. They play an important role in championing professional development and pedagogical excellence among the teaching fraternity.



Innovative approaches to teaching digital skills

Ms Teo Ying Hui, who teaches Chinese language in Changkat Changi Secondary School, uses digital tools to enhance her students' learning and to encourage them to use technology in a responsible way. She guides her students to use generative AI tools to refine their Chinese essays and encourages her students to critique the AI-generated responses. This way, students improve their command of the language, while also understanding that they must critically assess AI-generated content.

Similarly, Mrs Goh Xinyi from Fairfield Methodist School (Primary) uses the Short Answer Feedback Assistant, an AI-enabled feature in the Singapore Student Learning Space, to enhance teaching and learning in her classroom. This feature provides students with automatically generated feedback which teachers can build upon. While the feedback helps her students to improve, Mrs Goh also encourages her students to review the generated responses, emphasising that such AI-generated content may have limitations.

Both educators are passionate about making the most of technological advances, while remaining firmly grounded in the principles of good teaching and learning.

"We need to be well-informed about technology's uses to harness its benefits without getting lost in the whirlwind of rapid developments."

– Teo Ying Hui –



CHAPTER 4

Community



Community

Our efforts to build a trusted digital environment and grow opportunities for Singaporeans serve a higher purpose. Ultimately, they must help to shape a stronger, more united, and cohesive Singapore.

As we navigate the digital age, we must ensure that our Smart Nation benefits and uplifts all segments of society. We must uphold our shared values and aspirations, deepen our care for the people around us, and preserve our sense of togetherness. These are the features of the Singapore we hold dear and they must be strengthened in our Smart Nation.

We will find new ways to harness digital technology to unify our people. We will connect Singaporeans through shared interests, aspirations, and care for one another. We will use digital tools to deepen Singaporeans' engagement with our rich and diverse culture, arts, and heritage. We will also utilise technology to strengthen partnerships between government and citizens, as well as encourage citizens to co-create digital solutions that serve their communities and fellow Singaporeans.

To build a Smart Nation that keeps us together, we will:

A.



**Strengthen inclusion,
leaving no one behind**

B.



**Strengthen the unity
of our community**

C.



**Strengthen partnerships
with citizens**

A. Strengthening Inclusion, Leaving No One Behind

Everyone should have the opportunity to enrich their lives and benefit from digital technologies: whether it enables them to better communicate with friends and family, access essential services, or enjoy recreational activities.

Singapore must be a place where citizens are well-equipped with digital access and skills to embrace technologies. Digital products and services should be readily accessible and inclusive. Digital divides—where gaps in digital access and skills fuel social and economic inequalities—should not be allowed to persist.

We have supported over

62,000

low-income households with subsidised broadband and digital devices since 2020



i. Enhancing digital access for all

Ensuring widespread digital access is vital for empowering individuals and fostering inclusive economic growth in the modern world. In this regard, we have been investing significant efforts and resources, and made good progress over the years.

- ▶ Around 99% of our resident households are connected to the internet.³² By some measures, Singapore ranks first in the world for digital inclusion.³³
- ▶ We have supported over 62,000 low-income households with subsidised broadband and digital devices since 2020.
- ▶ In 2023, we launched the DigitalAccess@Home scheme to support lower-income households with subsidised broadband and digital devices through a simplified application process. Since then, the programme has supported more than 16,000 households. The Infocomm Media Development Authority's (IMDA) Mobile Access for Seniors scheme has also provided subsidised smartphone and mobile plans to over 13,000 seniors since its launch in 2020.

We will continue to identify gaps in digital access. In particular, we will monitor the challenges faced by low-income individuals and households, and introduce targeted interventions where needed.

³² Singapore Digital Society Report 2023, Infocomm Media Development Authority, 4 November 2023.

³³ Inclusive Internet Index 2022, Economist Impact, 2022.

ii. Equipping Singaporeans with digital skills for life

All citizens should have the opportunity to continually update the essential digital skills they need for everyday life. Singapore firmly believes in empowering our citizens with the knowledge, skills, and attitudes they require to safely and confidently navigate digital platforms and services.

- ▶ We set up the SG Digital Office (SDO) in 2020 to equip individuals and small businesses, including seniors and hawkers, with skills to participate in an increasingly digitalised environment. Singaporeans can learn digital skills at more than 30 SG Digital Community Hubs and about 200 roving counters across the island.

- ▶ We launched the Digital for Life movement to galvanise partners from the people, private, and public (3P) sectors to help Singaporeans embrace the skills, tools, and habits they need to thrive in a digital future. As of July 2024, the national movement has amassed over 300 partners from the 3P sectors and benefitted more than 400,000 Singaporeans, ranging from seniors to persons with disabilities.
- ▶ We developed the Digital Skills for Life (DSL) framework in 2024 to better equip Singaporeans with the essential digital skills they need to conduct daily tasks online.
- ▶ We conduct free programmes and workshops on a variety of technology-related topics, such as experimenting with emerging technologies like 3D printing at our public libraries.

The Digital for Life movement has benefitted

>400,000

Singaporeans, ranging from seniors to persons with disabilities



A Community Development Council workshop for seniors to learn digital skills.



Image Source: SG Digital Office



Learning to safely navigate online spaces at every age

Madam Cheong Poh Ching participated in a two-hour workshop organised by Singtel and SDO in June 2024 at Boon Lay Community Club, having attended such sessions with SDO for the past two years. The workshop covered topics like guarding against common scams and staying safe online. Madam Cheong found the session very beneficial as it equipped her with the essential skills to identify scams, recognise fake websites, and filter calls and SMSes from unknown numbers via the ScamShield mobile app.

"These digital skills have made my life more convenient. Now, I encourage my fellow seniors and bring my friends along to learn at the SDO."

– Cheong Poh Ching –

Ms Deena Sabrina bte Anwar, an Account Manager for Customer Success, Consumer Business Segment at Singtel Singapore, volunteered at the same series of workshops that Madam Cheong attended. During the workshop, she taught seniors how to navigate online spaces safely and explained the meaning of certain online terms used in both English and Malay to help seniors better understand them.

"The experience was a fulfilling and enriching one. It deepened my understanding of the challenges faced by some seniors as they adopt digital technologies. I look forward to volunteering more to help seniors stay safe online."

– Deena Sabrina bte Anwar –



Today, 68% of Singaporeans have basic or higher digital proficiency.³⁴ Nonetheless, some citizens across various age groups continue to find technology daunting.



We will strengthen our outreach efforts and partnerships to reach more Singaporeans and help them learn digital skills

with confidence. More than 30 partners have already come on board to help Singaporeans learn essential skills under the DSL framework. For example:

- ▶ We have partnered with the Council for Third Age to develop over 180 digital skills courses that are aligned with the DSL framework.
- ▶ Through the SDO@Workplace programme, we will reach more citizens at their workplaces by strengthening our collaboration with employers from various sectors including transport, food and beverage, healthcare, and more.

Apart from making everyday activities more convenient, digital skills can help us develop new hobbies and enrich our everyday experiences.



Finding joy in new hobbies

Lee Mei Ling, age 58, is a maker coach for the National Library Board's (NLB) MakeIT at Libraries, a makerspace that allows members of the public to try their hand at tech-based fabrication technologies such as 3D printing and laser digital cutters.

Mei Ling did not start out as a maker coach. She first decided to learn 3D printing to teach her sons a new skill. She found great joy in learning about 3D printing, especially when she realised that she could create beads for her Chinese knotting projects. She was even able to draw and print replacement parts to fix broken items at home—such as rubber stoppers on hangers or locks on storage boxes—rather than buying new ones.

Today, Mei Ling's skills have grown by leaps and bounds. She has created a full Chinese chess board, complete with chess pieces. She enjoys sharing her passion with others and has signed up to be a maker coach, teaching participants how to use the technologies at makerspaces.

“It is fun to pick up digital skills! People should not be daunted by them. Experimenting, getting things wrong, and tinkering with technology are part and parcel of the learning experience.”

– Lee Mei Ling –



³⁴ Digital Readiness Survey 2022, Ministry of Digital Development and Information, 2023. Basic or higher proficiency means having the ability to do basic tasks including accessing government services using Singpass, buying products or services online, using a smartphone to make contactless payment, setting a strong password for online accounts, enabling two-factor authentication, browsing or reading content on social media, searching for information on a topic online, and using instant messaging.

iii. Designing for accessibility

Digital products and services should be accessible, convenient, and easy for everyone to use, including seniors and persons with disabilities. We will also continue to keep key physical services available for those who are unable to use digital channels.

We will make our digital products more accessible and inclusive, as well as strengthen their usability, both within the government and in partnership with the private sector.³⁵

a. Government to lead by example

Singapore's approach is 'digital first', not 'digital only'.

- ▶ We will ensure there are adequate in-person touchpoints and physical alternatives available for Singaporeans who are still uncomfortable with navigating online services on their own.
- ▶ We will set up two more ServiceSG centres in Woodlands and Bukit Merah by end-2024, adding to the seven centres already in place.³⁶

The government also takes the accessibility, inclusivity, and usability of our products seriously. As outlined in the Enabling Masterplan 2030, we aim to make all high-traffic government websites accessible and are on track with 73% of websites accessible in 2023.³⁷

Improving accessibility with new developer tools



GovTech's A11y team has developed Purple A11y, an open-source web accessibility testing tool. It detects accessibility issues such as the level of colour contrast between text to ensure legibility for both visually impaired and elderly users. This allows software development teams to resolve identified bugs and improve the accessibility of their digital services.

Purple A11y is also starting to see usage worldwide beyond Singapore. Since July 2023, Purple A11y has scanned over 790,000 pages and discovered more than 11.5 million accessibility issues.



ServiceSG centres offer Singaporeans one-stop access to a wide range of government services.

³⁵ Accessibility and inclusivity refer to designing products and services that benefit all population segments, including the elderly and persons with disabilities. Usability refers to developing products and services that are simple to understand and easy to use.

³⁶ Existing ServiceSG centres are located at One Punggol, Our Tampines Hub, Bukit Canberra, The Frontier Community Club, Keat Hong Community Club, Nee Soon Central Community Club, and Kampong Chai Chee Community Hub.

³⁷ Enabling Masterplan 2030, Ministry of Social and Family Development, 17 August 2022.



Today, we have a set of Digital Service Standards (DSS) in place to help agencies design and develop digital services that are accessible, inclusive, and usable. It incorporates the Web Content Accessibility Guidelines, a set of globally recognised guidelines that relate to accessibility of web content. **We are embarking on a review of the DSS to keep it relevant.**

The Government Technology Agency of Singapore (GovTech) also conducts regular usability checks on frequently used services to help agencies identify areas for improvement. It offers tools to help agencies monitor and enhance the accessibility of their digital services. These tools include the Whole-of-Government Application Analytics, a centralised service that monitors and provides recommendations to raise the performance and usability of digital services, and Purple A11y, which helps agencies identify and rectify accessibility issues on their websites.

b. Deepening engagement with the private sector

Private organisations play an important role in developing inclusive products and services for citizens. Commercial digital services and products like online banking, e-commerce, ride-hailing, and telemedicine are commonly used by Singaporeans in their daily lives.



We will build on existing efforts, such as GovTech's Co-Creation Lab and SG Enable's initiatives, to engage with private sector and community partners to raise awareness about accessibility and facilitate knowledge sharing. We will also work with the private sector to co-create and pilot solutions together, strengthening our ecosystem of accessible digital products and services.

Designing inclusive digital products together



The inaugural Co-Creation Lab, organised by GovTech's A11y team, was launched during A11y Week in May 2024. It brought together representatives from government agencies, the private sector such as participants from banks and e-commerce firms, as well as community users—especially seniors and persons with disabilities—to better understand accessibility-related challenges.

Users shared the challenges they faced while accessing digital products and services to conduct their daily activities. These included issues such as needing a longer time to complete two-factor authentication for banking apps and struggling with multiple steps required to log in and access services. Product designers were also encouraged to reflect on how they might make their products more inclusive.



Image Source: Government Technology Agency



Helping every voice be heard

Technology holds tremendous potential for enabling persons with disabilities to carry out daily activities with greater convenience and ease.

In 2023, Google partnered with the Movement for the Intellectually Disabled of Singapore (MINDS) to beta test Project Relate, a speech-to-text app for people with non-standard speech. The app allows users to submit samples of their voice to create a personalised speech recognition model that more accurately understands their speech. For example, the app can transcribe an individual's speech to text and repeat it, overcoming communication barriers faced by individuals with atypical speech.

Sharing his ideas with the people around him can often be a challenge for Allan Cai, who always had to repeat himself multiple times to be understood by others. The Project Relate app has helped him articulate his thoughts clearly. He now finds that friends and colleagues are impressed by his insights on current affairs. With an app like Project Relate, Allan looks forward to a future that will be more inclusive for people with non-standard speech.

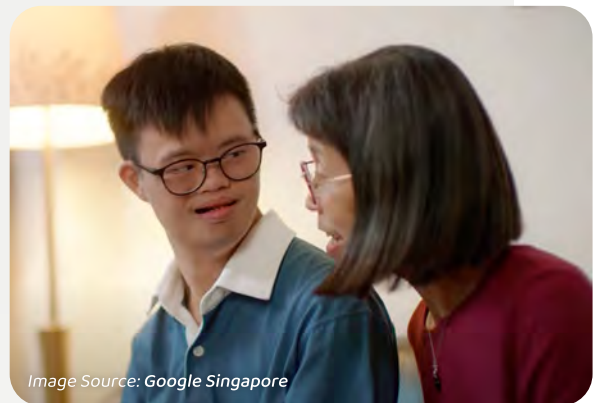


Image Source: Google Singapore

Ang Wan Qi, a customer engineer at Google Cloud, was instrumental in rolling out the Project Relate app to selected testers in 2023. She was the bridge between the product team in the United States of America and local users like Allan. Given the app's focus on American-centric language and phrases, Wan Qi helped Allan to include everyday Singaporean greetings and phrases in the app. She has also led training sessions for MINDS, enabling local caregivers and volunteers to support persons with non-standard speech in using the Project Relate app.



Following the success of the beta testing phase, Google will be launching the Project Relate app in the Singapore Google Play Store this year. Google is also working with MINDS and other organisations to enable more people to benefit from the app.

B. Strengthening the Unity of Our Community

Technology can help us traverse distance and time and break down barriers between people. We believe in its power to bring together Singaporeans from all walks of life, deepen our understanding of one another, and enable our shared culture to thrive.

But it can also derail our sense of community. Countries all over the world are grappling with digital developments that undermine cohesion, such as the formation of echo chambers and increasingly polarised discourse. We must be a society where Singaporeans work together to nurture a positive, civic-minded culture online, and where digital technology is used to edify rather than denigrate, unify rather than divide.



Image Source: National Museum of Singapore

Reunion space at the National Museum of Singapore.

i. Connecting more Singaporeans

Digital technologies are opening new pathways for citizens to connect with our culture, arts, and heritage. Through innovative platforms and virtual experiences, we can explore and engage with our rich cultural legacy, ensuring that our traditions and creative expressions continue to thrive.

In the arts and heritage sector, digital technology has helped enrich Singaporeans' experiences of our cultures and traditions, such as through gamifying heritage trails and weaving technology into the curation of exhibitions.

- ▶ The Reunion space exhibit in the National Museum of Singapore allows seniors to curate their own virtual exhibitions with digital replicas of artefacts like Singer sewing machines and vintage chairs, creating opportunities for them to reconnect with their past, and share memories with family and friends.

We have also used digital platforms to bring the community together. These tools help us stay connected, share experiences, and collectively work towards causes that strengthen our bonds and create positive change.

- ▶ Giving.sg, a platform by the National Volunteer and Philanthropy Centre, has raised over S\$600 million for good causes and facilitated volunteering opportunities for over 300,000 people since 2010.



We will continue to launch and support more initiatives that re-imagine and reinvigorate how we connect with one another and our culture using technology.





Keeping our *kampung* spirit alive in a digital age

In Singapore, '*kampung* spirit' is synonymous with a strong sense of community and togetherness. GovTech's Open Government Products (OGP) engineers have built a digital platform named KampungSpirit that seeks to keep this spirit alive. KampungSpirit helps social workers source for essential goods and services to help the needy.

Mabel Ng, a social worker at one of Allkin Singapore's Family Service Centres, has used the platform on several occasions to gather essential resources such as washing machines and wheelchairs for her clients.

"I saw first-hand the generosity and heart for the community through the app. The process of matching clients to essential resources can be quite tedious so having this digital option has really smoothened the process and widened our pool of donors."

– Mabel Ng –



Exploring Singapore's heritage in 3D

The National Heritage Board is developing a National 3D and Digital Heritage Repository to document Singapore's cultural heritage using 3D, geospatial, and digital technologies.

The pilot repository will make high-quality cultural and heritage content and assets more accessible, enabling more innovative approaches to heritage research and education. For example, students can explore a 3D replica of the Singapore Stone,³⁸ and other Singaporeans who may not be able to make physical trips to our museums can experience our artefacts up close. The repository will also help establish Singapore as a centre of excellence for 3D and digital heritage.



38 The Singapore Stone is a sandstone boulder discovered at the mouth of the Singapore River. It is believed to date between the 10th to 14th centuries.

The archives come to life with generative artificial intelligence (AI)



NLB has developed ChatBook, a generative AI-powered prototype chat service where users 'converse' with a book. The first publicly available iteration of the prototype draws from the rich resources of the National Archives of Singapore and the authorised biography of founding father S. Rajaratnam to help users learn more about his life and contributions.



Image Source: National Library Board

ii. Growing a culture of positivity and civic-mindedness online

Digital platforms can undermine our sense of belonging by emphasising divisions across and within communities. According to a 2021 YouGov survey, 64% of respondents indicated that they had observed increased polarisation of views online in Singapore in the previous five years.³⁹

This is cause for concern especially if viewpoints become increasingly entrenched across ideological or even racial and religious lines. This will undermine our capacity as a country to build consensus, shared norms, and a path forward together.

We must be a society where Singaporeans help to nurture a civic-minded, positive community online and protect our social cohesion against eroding effects. There are many issues to grapple with, including how to build social norms together in an internet that is global and borderless by nature.



We will study these issues carefully and explore ways to nurture a positive online space.

Globally, there are examples of organisations that have designed collaborative online platforms that nudge people towards civic behaviour. For instance, mechanisms that allow users to upvote content that attempts to reconcile different views, rather than incendiary content that usually attracts more engagement online.

Some young Singaporeans have already stepped up to encourage more civic-minded behaviour online.

³⁹ Survey commissioned by Channel NewsAsia and reported in "IN FOCUS: 'Boomer, snowflake, oppie, pappie' - unpacking the growing social media polarisation in Singapore", Channel NewsAsia, 7 August 2021.



Nurturing young digital citizens

In 2023, the School of Science and Technology started a Digital Citizenship (DC) Board that is entirely student-led. Students on the DC Board help their peers and juniors to grow their cyber wellness skills, through initiatives like facilitating discussions on the ethical considerations behind using technology. They understand that youth today must grapple with complex topics online, such as balancing the right to speak freely with respecting others' feelings. They also need to navigate grey, often undefined areas, including the norms and etiquette surrounding the use of technology.

To explore these complex issues, students on the DC Board initiate and conduct Digital Inquiry research projects. They also embark on capability building efforts, such as equipping cyber wellness ambassadors from primary schools to grow their facilitation skills so that they too can positively shape their peers' views and behaviours regarding online issues.

The nuanced, thoughtful discussion led by such youth advocates add a meaningful dimension to our Smart Nation vision, as we work together to nurture more positive, civic-minded communities online.

“With digital issues being such a big part of students' lives, it is important to empower them, get them to think deeply about these issues, and have conversations among themselves”

- Raymond Chng -
Teacher-in-charge of the DC Board



C. Strengthening Partnerships with Citizens

We want to engage citizens to create meaningful change and play their part to shape the next chapter of Singapore's story, through digital technology. Smart Nation is ultimately a whole-of-nation effort.

Technology can also be used to strengthen the partnership between citizens and government, opening channels for citizens to share feedback and co-create ideas. We can also grow a community where everyone is empowered to build digital solutions to tackle pressing community needs.



We will create opportunities for interested citizens to work with the government to co-create innovative digital solutions to address issues and solve problems that they care about.



i. Citizens as builders

We want to partner citizens to identify and solve problems using digital technology. By coming together and contributing solutions or insights on problems faced by the community, we can develop meaningful digital tools that improve the lives of our fellow citizens.

We will create opportunities for interested citizens to work with the government to co-create innovative digital solutions to address issues and solve problems that they care about. This collaborative approach nurtures citizens' sense of ownership and taps their insights for developing creative digital solutions that serve the public good.





Innovation from the ground up

Build For Good (BFG) is a citizen engagement programme that was started by OGP in 2023. BFG brings citizens together through a month-long hackathon to identify a problem they care about, ideate solutions, build prototypes, and test them with users. Participants are provided with mentorship, training, and resources, with winning teams embarking on an eight-week accelerator programme where they receive funding to refine and potentially scale their products. Government entities such as the Singapore Partnerships Office will also work with OGP to conduct hackathons to engage citizens on issues close to their hearts.

In May 2024, OGP held its second BFG focused on sustainability in partnership with the SG Eco Fund. Eighty builders formed 19 teams to build solutions that addressed problems in the environment sector. Through the accelerator programme, five winning teams secured partnerships with government agencies, private companies, and non-governmental organisations to run their pilots.

One of the winning solutions was RemediSG, an online platform that seeks to redistribute excess unused medication to other healthcare providers and minimise waste. Audrey Kon, a nurse by profession and member of the winning team, noted that participating in the BFG hackathon was an invigorating experience that empowered her to apply her tech know-how to solve a unique and challenging problem within the healthcare sector. The resources provided to her team helped them transform their ideas into a tangible prototype that is now being piloted with two healthcare organisations.

“Build For Good equipped me with new perspectives that value-added to my experience in the healthcare sector. It inspired me and my team to ideate a solution and bring it to fruition. I hope the product will encourage healthcare organisations to work towards sustainable practices.”

– Audrey Kon –



ii. Expanding platforms for citizen engagement

Digital platforms are also a key channel for government agencies to gather ideas and feedback from citizens on government policies. We will continue to use these platforms to engage our citizens on government policies, so that citizens' needs and aspirations can be better reflected.

- ▶ CrowdTaskSG is an online platform that allows government agencies to gather feedback on policies and services. The platform enhances engagement by gamifying the experience for citizens. For instance, it allows them to collect 'coins' that can be used to redeem cash rewards at participating merchants when they give feedback through opinion surveys.
- ▶ We have set up the Singapore Government Partnerships Portal for citizens to share their partnership proposals with the government.



We regularly gather feedback on government digital products and services to keep citizens' needs at the heart of the development process.



We regularly gather feedback on government digital products and services to keep citizens' needs at the heart of the development process. Feedback from actual users makes our digital offerings more responsive to the changing needs of our citizens and ensures user-friendliness.

- ▶ GovTech's Tech Kaki citizen community consists of almost 6,000 citizens from diverse backgrounds who participate in focus group discussions, user testing, and surveys to provide feedback on digital products and services. A user research study conducted through this community provided insights on the redesign of the Singpass app consent screen to help citizens better verify the services they are logging into.



GovTech's Tech Kaki citizen community.

CHAPTER 5

Harnessing Technology for Global Good



Harnessing Technology for Global Good

The digital realm crosses physical borders, presenting challenges and opportunities that necessitate international cooperation. For example, complex and rapidly evolving digital technologies require governments to collaborate on regulatory and security approaches.

When countries and institutions share knowledge and experiences the entire world benefits, especially when we work together to develop practical solutions, set interoperable standards, and shape new norms.

Singapore remains committed to working with our international network of friends and partners to harness technology as a force for global good. We are firm believers in fostering an inclusive approach to capacity building and rule-making. We collaborate with international partners to tackle global challenges and will readily do our part to build bridges in support of the Global Digital Compact.⁴⁰



⁴⁰ The Global Digital Compact is an annex to the Pact of the Future, a document which articulates a renewed vision for the United Nations ahead of the 10th anniversary of the Sustainable Development Goals in 2025. The Global Digital Compact annex outlines principles, objectives, and actions for advancing an open, free, secure, and human-centred digital future.

Harnessing Technology for Global Good

A. Fostering an Inclusive International Community for Digital Development



We believe in the value of the international community coming together to shape how technology develops, and build consensus around international norms and rules. These conversations and the digital ecosystems they result in should be inclusive, reflecting views and concerns from a wide range of countries and stakeholders.

Singapore has been a strong advocate for more inclusive approaches toward technology development and governance. We participate actively in multilateral and regional fora, and contribute to global discourse on key digital issues.

- ▶ We work within the Association of Southeast Asian Nations (ASEAN) to build a trusted and inclusive digital ecosystem, so that the region prospers and benefits from technological developments.
 - » Under Singapore's Chairmanship of the ASEAN Digital Ministers' Meeting in 2024, we developed frameworks to govern emerging technologies like artificial intelligence (AI). The ASEAN Guide on AI Governance and Ethics, launched in February 2024, was the first concerted attempt by ASEAN to put forward common guidelines supporting AI adoption and protecting our societies against the irresponsible or unethical use of AI.
 - » We launched the European Union (EU)-ASEAN Joint Guide on Model Contractual Clauses for International Data Transfers which provides greater clarity on the management of cross-border data flows for businesses operating in the EU and ASEAN.
- ▶ In 2022, a digital pillar was introduced under the Forum of Small States (FOSS), which Singapore convenes.⁴¹ The Digital FOSS provides a platform for small states to exchange views and best practices on digital issues, build capacity, and ensure that small states' voices are heard in platforms and fora that shape our digital future. We have worked with Rwanda to develop an AI Playbook for Small States, which aims to share key considerations and best practices for AI development, governance, and use with policymakers.
- ▶ We participate actively in multi-stakeholder platforms and initiatives including the AI Safety Summit series, the Global Partnership on AI, the World Economic Forum's AI Governance Alliance, and the Hiroshima AI Process Friends Group. This is to shape a more inclusive, practical, and rules-based global environment for AI.



Singapore has been a strong advocate for more inclusive approaches toward technology development and governance. We participate actively in multilateral and regional fora, and contribute to global discourse on key digital issues.



Image Source: Infocomm Media Development Authority

Digital FOSS Fellowship Programme, 2024.

⁴¹ The FOSS comprises 108 countries.



Image Source: Government Technology Agency

Digital Government Exchange, 2024.



Image Source: Infocomm Media Development Authority

Asia Tech x Singapore, 2024.



Singapore Conference on AI for the Global Good, 2023.

In addition, we bring thought leaders from around the world to Singapore for impactful discussions about the digital opportunities and challenges of our time.

- ▶ Since 2016, we have convened the annual Digital Government Exchange as an international platform for government Chief Information Officers and public sector leaders to share experiences and explore collaboration opportunities.
- ▶ We established Asia Tech x Singapore in 2021 as a platform for global thought leaders to enhance mutual understanding on digital issues. The platform has grown steadily and become one of the region's most prominent technology fora.
- ▶ In 2023, we organised the inaugural Singapore Conference on AI for the Global Good in partnership with Topos Institute. It was attended by over 40 local and overseas AI experts who explored critical questions related to AI, including how we could best develop AI to keep humanity flourishing.



Harnessing Technology for Global Good

B. Building Bridges



Countries have a common interest in technological advancement and progress. It is crucial that we invest in our global digital commons, collaborate, and promote interoperability. Singapore contributes to these efforts on multiple fronts.

We are committed to deepening our bilateral and multilateral partnerships, including through facilitating trade, innovation, and information sharing.

- ▶ We have established multiple Digital Economy Agreements (DEA), including the plurilateral Digital Economy Partnership Agreement (DEPA), to facilitate seamless cross-border trade and innovation between digital economies.

- » DEPA is a first-of-its-kind agreement that establishes rules for digital trade and fosters cooperation between economies in emerging digital areas. It was jointly initiated by Singapore, Chile, and New Zealand in 2020; the Republic of Korea came on board in 2024.

- » To date, our DEA partners include Australia, the United Kingdom (UK), the Republic of Korea, and the EU.

- ▶ We are working with key global partners to deepen mutual understanding and information sharing, such as with the United States of America (US) through the US-Singapore Critical and Emerging Technologies Dialogue, as well as with China through the Singapore-China Digital Policy Dialogue.

We help to shape and align international standards and norms, with a view towards promoting interoperability.

- ▶ In 2023, we completed a joint mapping exercise between the Infocomm Media Development Authority's AI Verify and the US National Institute of Standards and Technology's Risk Management Framework. This is an important step towards the harmonisation of international AI governance frameworks. We have agreed to take our work further by mapping our respective frameworks on generative AI. With China, we have agreed to promote common baseline standards to facilitate cross-border flows of commercial data.

- ▶ We are laying the groundwork for the safe use of quantum technology, co-leading the first standardisation of the Quantum Key Distribution protocol framework at the International Telecommunication Union's Telecommunication Standardisation Sector, together with Japan.

- ▶ In 2020, we launched the Cybersecurity Labelling Scheme (CLS) for consumer smart devices to improve cybersecurity for the internet of things and raise overall cyber hygiene levels. The CLS is among the first of its kind globally and has garnered international attention. Labels issued under the CLS are recognised by Finland, Germany, and the Connectivity Standards Alliance.



C. Tackling Global Digital Security Challenges



Governments around the world have become familiar with the challenges posed by digital technologies to their citizens' safety and wellbeing. Singapore has cultivated strong partnerships with other nations to tackle cybercrime and cybersecurity threats, given the transnational nature of these challenges.

- ▶ We launched the ASEAN-Singapore Cybersecurity Centre of Excellence in 2019. It deepens and enhances the region's cybersecurity posture, and capacity to respond to cyber threats and challenges, by exchanging timely information, best practices, and capacity building initiatives.
- ▶ Singapore, alongside countries such as the US, the UK, and Germany, is also among the pioneering members of the Counter Ransomware Initiative (CRI), a global coalition collectively addressing the threat of ransomware. Together with the UK, Singapore co-chairs the CRI Policy Pillar to strengthen international collaboration to disrupt the global ransomware ecosystem.
- ▶ We collaborate with international partners to enhance cross-border cooperation and fight scams. Our partners include the US, Malaysia, Australia, New Zealand, and Hong Kong, China.
- ▶ Singapore chairs the ongoing five-year United Nations (UN) Open-Ended Working Group (OEWG) on the Security of and in the Use of Information and Communications Technologies (2021–25). The OEWG helps to shape cybersecurity norms in the UN and enables states to engage in discussions to support and promote an open, secure, and interoperable ICT environment.
- ▶ We convened the inaugural Singapore International Cyber Week (SICW) in 2016 as a platform for policymakers, industry leaders, and academia to exchange views on top-of-mind cybersecurity issues and strengthen cybersecurity cooperation. The SICW has since become Asia-Pacific's most established cybersecurity event. In recent years, it has brought together senior leaders from the US, China, Russia, the UK, France, the EU, Ukraine, and across Asia to engage in robust discussions on challenges confronting our collective cyberspace.



Image Source: Cyber Security Agency of Singapore

Singapore International Cyber Week, 2022.



Image Source: Cyber Security Agency of Singapore

Harnessing Technology for Global Good

D. Contributing to Capacity Building



Singapore is doing our part to close the global digital divide by building capacity at the regional and global levels. We readily share our expertise in using digital solutions to improve citizens' lives while learning from peers and experts worldwide.

- ▶ We partner with the UN Development Programme through its Global Centre for Technology, Innovation, and Sustainable Development in Singapore to identify and co-create solutions for sustainable development. These include supporting their work on a Digital Inclusion Playbook which provides a foundational set of frameworks, policy recommendations, and global case studies to help other countries become more digitally inclusive.
- ▶ We are a contributing member of the International Telecommunication Union's Partner2Connect Digital Coalition and have pledged S\$10 million to help the international community harness the potential of digital technologies, in line with the UN Sustainable Development Goals.
- ▶ Singapore joined the UN-endorsed Digital Public Goods Alliance in 2024. We have made open-source versions of high-impact Singapore government products like FormSG, which helps public officers create digital forms within minutes. Other nations have used these products to meet critical needs in areas such as information accessibility and trustworthy communication. We have collaborated with countries like Indonesia, Cambodia, and Sri Lanka to deploy our open-source products.
- ▶ We run capacity building programmes in partnership with other countries, such as the US-Singapore Third Country Training Programme for officials from Southeast Asia.



We have made open-source versions of high-impact Singapore government products like FormSG, which helps public officers create digital forms within minutes. Other nations have used these products to meet critical needs in areas such as information accessibility and trustworthy communication.



Conclusion

The goals and initiatives outlined in this document illustrate what Smart Nation 2.0 seeks to achieve. Our focus is not on technology per se. We care much more about Singaporeans, our communities, and moving forward together as a nation.

Our strategies help establish important foundations for a Smart Nation that we can trust, helps us grow, and keeps us together.

We will be a Smart Nation where:

1.



We harness technology to raise our citizens' quality of life

2.



Singaporeans can go online with confidence, knowing that their safety is not compromised

3.



Technology empowers our people to learn and excel in better ways

4.



Technology brings us together and strengthens our society

5.



We foster an inclusive international community for digital development

Conclusion

We will be a Smart Nation where:

- ▶ **We harness technology to raise our citizens' quality of life.**
 - ▶ **Singaporeans can go online with confidence, knowing that their safety is not compromised.**
 - » Citizens and enterprises can depend on secure, resilient digital infrastructure. Our people and organisations are well-prepared to respond to and recover from disruptions if they occur.
 - » Harmful online activities are detected and disrupted quickly. Victims of online harms have access to effective help and redress as Singapore continues to adapt our laws, systems, and capabilities with speed and agility.
 - » Our community works together to build trusted digital spaces conducive to our wellbeing, especially for our children and youth. Our citizens are empowered with mindsets, skills, and tools to use technology responsibly.
 - ▶ **Technology empowers our people to learn and excel in better ways.**
 - » Our digital infrastructure continues to be robust, resilient, and future-proof. Singapore continues to extend our frontiers with artificial intelligence (AI), harnessing its transformative potential while creating a safe, trustworthy environment for AI development and deployment.
 - » Growth and progress are shared widely with our workers and enterprises who are empowered to adapt, succeed, and augment their capabilities with technology.
 - » Our youth are curious, confident, and creative users of technology who are poised to thrive in a digital age.
 - ▶ **Technology brings us together and strengthens our society.**
 - » All Singaporeans can enrich their lives and benefit from technology regardless of their circumstances. Government and private sector organisations design digital products and services that are accessible, inclusive, and user-friendly.
 - » Digital technologies unify rather than divide. They bring together Singaporeans from all walks of life, deepening our understanding of one another and enabling our culture to thrive.
 - » Technology enables citizens to work in closer partnership with government, facilitating the active sharing of ideas, feedback, and co-creation of digital solutions that serve the community.
 - ▶ **Singapore continues to fly our flag high, working together with our friends and partners to foster an inclusive international community for digital development. We tackle common challenges together, help to develop interoperable standards and norms, and shape technology for the good of Singapore and the world.**
-



**We will be
a Smart Nation
that cares for all
Singaporeans,
giving everyone the
opportunity to thrive in
our digital future.**



Frameworks and Blueprints

In support of Singapore's Smart Nation vision

Singapore National AI Strategy 2.0

Outlines our vision for Singapore as a place where AI is harnessed as a force for good to uplift and empower our people and businesses.

Digital Readiness Blueprint

Sets out our goals and recommendations to help all Singaporeans use and benefit from technology in their everyday lives.

Digital Connectivity Blueprint

Sets out our strategic priorities and new frontiers to ensure that our digital infrastructure remains world-class and future ready.

Digital Government Blueprint

Details key performance indicators to measure the progress of our government's digitalisation efforts.

Digital Enterprise Blueprint

Sets out our ambitions and focus areas to empower enterprises and workers in the next bound of digitalisation.

Singapore Cybersecurity Strategy

Outlines our goals and approaches to strengthen digital security and resilience and enable a safer cyberspace.

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